

LIGHT

■ Light

Light is a form of energy which excites our sense of sight.

- **Sources of light :** During the day, the primary source of light is the Sun. Other common sources are flames, electric bulbs, tube lights (fluorescent tubes), compact fluorescent lamps (CFLs) and light emitting diodes (LEDs).

■ The nature of light

Light consists of electromagnetic waves (non-mechanical waves) which do not contain any particles and they do not require any material medium for their propagation.

- The speed of light depends on the nature of medium. In vacuum or free space, they travel fastest and their speed is 3×10^8 m/s. In air, the speed is almost same (marginally less), whereas, in glass or water, it reduces considerably.
- The wavelength range of visible light is about 4×10^{-7} m to 7×10^{-7} m.
- Though light does not contain particles, but it carry energy and momentum. Light is supposed to have massless particles or units called 'photons'.
- Light travels in a straight line in vacuum or in a homogeneous transparent medium.
- Light consists of electromagnetic waves which are transverse in nature.

■ How electromagnetic waves are produced ?

An electromagnetic wave is produced by an oscillating charge which produces mutually perpendicular oscillating electric and magnetic fields. The direction of propagation of electromagnetic wave is perpendicular to both electric and magnetic field respectively. Hence, it is transverse in nature.

- Electromagnetic waves are able to travel through a vacuum - a region without any matter. Mechanical waves require a medium in order to propagate from one location to another.

■ The electromagnetic spectrum

Electromagnetic waves exist with an very large range of frequencies. This continuous range of frequencies is known as the **electromagnetic spectrum**.

Gamma rays	X rays	Ultraviolet rays	Visible light	Infrared rays	Micro waves	Radio waves
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Decreasing frequency, increasing wavelength


- All regions of the electromagnetic spectrum have the same speed in vacuum. The speed of an electromagnetic wave in vacuum does not depend on its frequency and wavelength.
- **Visible light spectrum :** Our eyes are sensitive to only a very narrow band of electromagnetic spectrum. This narrow band of wavelengths is an aid through which humans can see and it is called **visible light spectrum**.

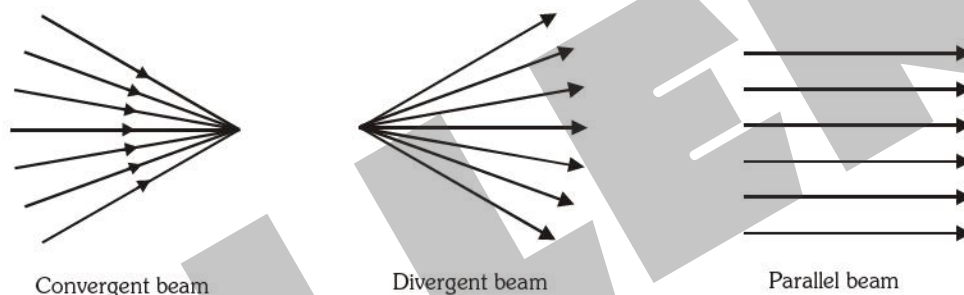
Violet	Indigo	Blue	Green	Yellow	Orange	Red
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Decreasing frequency, increasing wavelength


■ Objects

- **Luminous objects :** The objects which emit their own light are called '**luminous objects**'.
 ▶ **Examples :** Sun and other stars, lamp, bulb, tube light, candle flame, etc.
- **Non-luminous objects :** The objects which do not emit their own light but only reflect or scatter the light which falls on them are called '**non-luminous**' objects (or **illuminated objects**).
 ▶ **Examples :** Table, chair, animals, plants, planets, satellites, moon, etc.
- When the light falls on a non-luminous object, it reflects a part of the light towards us. When this light reaches our eyes, we could see such an object. A luminous object emits its own light. When this light reaches our eyes, we could see a luminous objects. Thus, for you to see an object, it must reflect or emit some light that reaches your eyes.

- **Opaque objects** : An object is said to be 'opaque' if it does not allow any light to pass through it. An opaque object reflects a part of the light falling on it and remaining part is absorbed by it.
- **Transparent objects** : An object is said to be 'transparent' if it transmits most of the light falling on it. Only a very small part of light may be absorbed or reflected from it. We can see clearly an object present behind the transparent object. *e.g.* a plain glass slab is a transparent object (or medium).
- **Translucent objects** : An object is said to be 'translucent' if it transmits a part of the light falling on it that is not sufficient to see clearly an object present behind it.
 - ▶ For a highly polished surface or mirror, most of the light falling on it gets reflected. For a transparent medium like glass or water, most of the light falling on it gets transmitted (refracted).
 - ▶ When light falls on a surface or a boundary separating two media, usually a part is reflected, a part is transmitted (refracted) and a part is absorbed.
- **Ray of light** : A ray of light is the direction in which light travels.
 - ▶ A ray of light is an idealization. In reality, it is a narrow beam of light which is made up of several rays. For simplicity, we use the term 'ray' for a narrow beam of light.
- **Beam of light** : A bundle of light rays is called **beam of light (or light beam)**.
- **Convergent beam** : A beam of light in which all the rays move towards a single point is called **convergent beam**.
- **Divergent beam** : A beam of light in which all the rays emerge out from a single point is called **divergent beam**.
- **Parallel beam** : A beam of light in which all the rays are parallel to each other is called **parallel beam**.



Reflection of light

Reflection of light is the process in which light rays meeting the boundary between two media 'bounces back', to stay in the first medium.

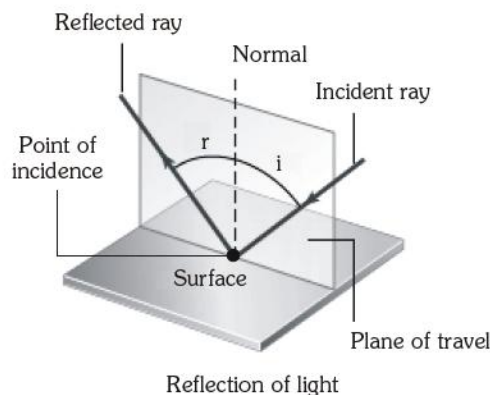
- The process of bouncing back of light rays which fall on the surface of an object into the same medium is called **reflection of light**.
- On reflection of light from a surface, the velocity, wavelength and frequency of light does not change. This is because the light stay in the same medium. But, amplitude and intensity of reflected ray is slightly less than that of incident ray as some part of energy is absorbed at the surface.

Laws of reflection

- ▶ **First law** : The incident ray, the reflected ray and the normal at the point of incidence, all lie in the same plane.
- ▶ **Second law** : The angle of incidence is equal to the angle of reflection.

$$\angle i = \angle r$$

- **Incident ray** : The ray of light which falls on the mirror surface is called **incident ray**.
- **Reflected ray** : The ray of light which is sent back by the mirror is called 'reflected ray'.
- **Point of incidence** : The point at which the incident ray falls on the mirror is called 'point of incidence'.
- **Normal** : A line perpendicular to the surface of mirror passing through the point of incidence is called 'normal'.

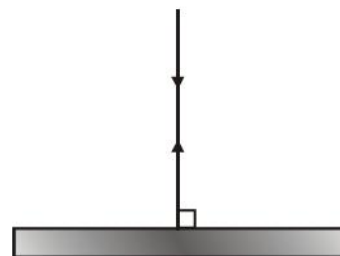


- **Angle of incidence** : The angle made by incident ray with the normal at the point of incidence is called 'angle of incidence'.
- **Angle of reflection** : The angle made by reflected ray with the normal at the point of incidence is called 'angle of reflection'.
 - ▶ The laws of reflection holds good for all kinds of waves. These laws are applicable to plane as well as curved surfaces.
 - ▶ When a light ray fall perpendicular on the surface of a mirror, it reverses its path on reflection (see fig.). That is, it exactly retraces its path because angle of incidence and angle of reflection both are equal to zero.

■ Regular reflection (specular reflection)

If a parallel beam of light is incident on a smooth plane mirror, the reflected beam is also a parallel beam. This reflection is called 'regular reflection'.

Reason : The light rays strike the smooth surface at same angles of incidence. This is because all the normals are parallel to each other on the smooth surface. Thus, the light rays get reflected at same angles of reflections. That is why, the rays of reflected beam remains parallel to each other.

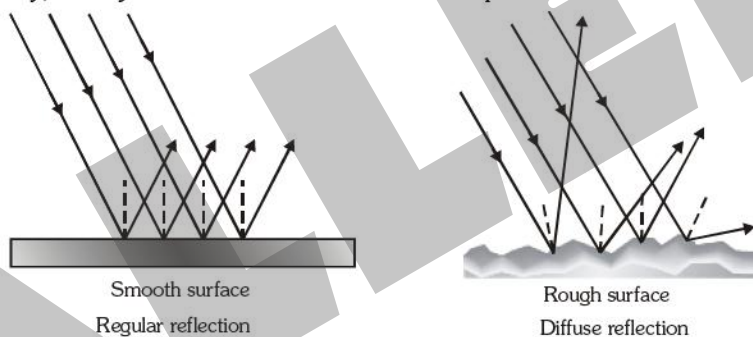


A Ray falling normally on the mirror retraces its path

■ Diffuse reflection (irregular reflection)

If a parallel beam of light is incident on a rough surface or mirror, the rays of light becomes non-parallel to each other. Such a reflection is called 'diffuse reflection'.

Reason : The light rays strike the rough surface at many different angles of incidence. This is because all the normals are not parallel to each other on the rough surface. Thus, the light rays get reflected at many different angles. That is why, the rays of reflected beam become non-parallel to each other.



■ Reflection from plane mirrors

A mirror is a highly polished surface used to reflect the light falling on it. Mirrors are usually made by depositing a thin layer of silver metal on one side of a plane glass sheet.

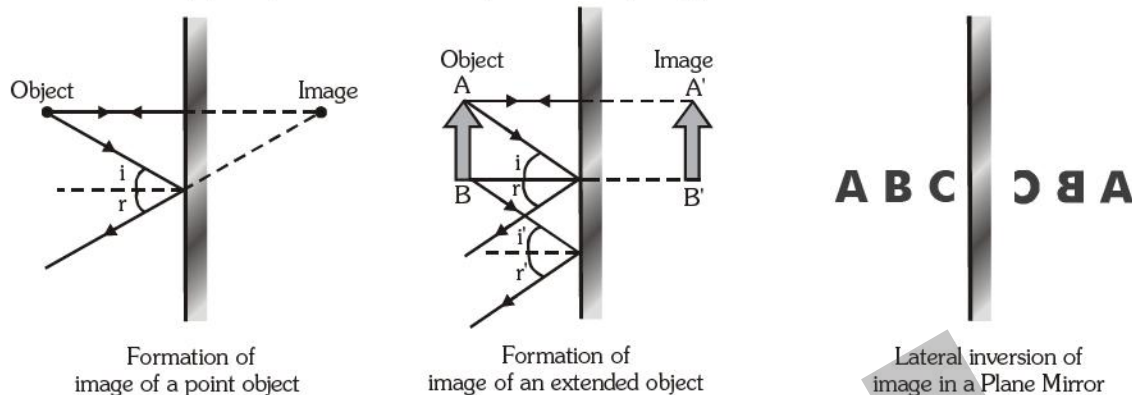
- **Object** : Anything which gives out light rays either of its own or due to reflection is called an 'object'.
 - ▶ **Point object** : An object whose dimensions are negligibly small with respect to the reflecting surface is called 'point object'.
 - ▶ **Extended object** : An object whose dimensions are quite large is called 'extended object'.

Point Objects	•	} Symbols used in ray diagrams
Extended Objects	↑	

- **Image** : An image of an object is formed when light rays coming from the object meet or appear to meet at a point after reflection from a mirror or refraction from a lens.
- **Real image** : A real image is one formed when the light rays actually meet at a point and which can be obtained on a screen.
- **Virtual image** : A virtual image is one formed when the rays do not actually meet at a point but they appear to meet at a point. Such images can not be obtained on the screen.
- **Image formed by a plane mirror** : The properties of image formed by a plane mirrors are,
 - ▶ The image is virtual and erect.
 - ▶ The distance of image from mirror is equal to distance of object from mirror.
 - ▶ The size of image is exactly equal to the size of object.
 - ▶ The image is laterally inverted.

● Lateral inversion

When an asymmetric object is placed in front of a plane mirror, then the right side of the object appears to be the left side of image and the left side of the object appears to be the right side of its image. This change of sides of an object seen in the image is called 'left - right inversion' or 'lateral inversion'. The image is inverted side ways, thus, also called 'side ways inversion' (see fig.).



- **Angle of deviation (δ)** : The angle through which the incident ray of light is deviated from its original path when it is reflected from a mirror is called 'angle of deviation'.

- The angle between incident ray and reflected ray is called 'angle of deviation'.
 - ▶ In general, if a ray is reflected (or refracted) many times, then the angle between the initial ray (incident ray) and the final ray is called 'angle of deviation'.

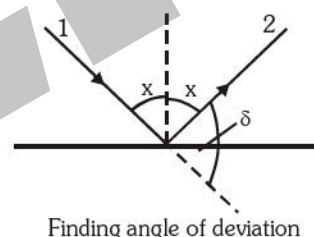
● Angle of deviation for reflection from a single mirror :

Let an incident ray 1 falls on a plane mirror at an angle of incidence x , then by the law of reflection, the angle of reflection is also x (see fig.).

Now, we can find angle of deviation (δ) from the figure,

$$x + x + \delta = 180^\circ \text{ (angle of a straight line is } 180^\circ)$$

$$\text{or } 2x + \delta = 180^\circ \quad \text{or} \quad \delta = 180^\circ - 2x$$



● Angle of deviation for reflections from two plane mirrors:

Let us consider two plane mirrors inclined at a certain angle ' θ '. Let an incident ray 1 falls on the mirror M_1 at an angle of incidence ' x ', then by the law of reflection, the angle of reflection is also x (see fig.). The ray reflected from the mirror M_1 (ray 2) acts as an incident ray for the mirror M_2 . Let the ray 2 falls on the mirror M_2 at an angle y , then by the law of reflection, the angle of reflection is also y . Let the ray reflected from mirror M_2 be ray 3.

Total angle of deviation = angle of deviation for M_1 + angle of deviation for M_2

$$\text{or } \delta = \delta_1 + \delta_2$$

$$\text{or } \delta = (180^\circ - 2x) + (180^\circ - 2y)$$

$$\text{or } \delta = 360^\circ - 2(x + y) \text{ ---- (1)}$$

From fig.6, we can see that,

$$\theta + (90^\circ - x) + (90^\circ - y) = 180^\circ$$

($\angle$ sum property of a Δ)

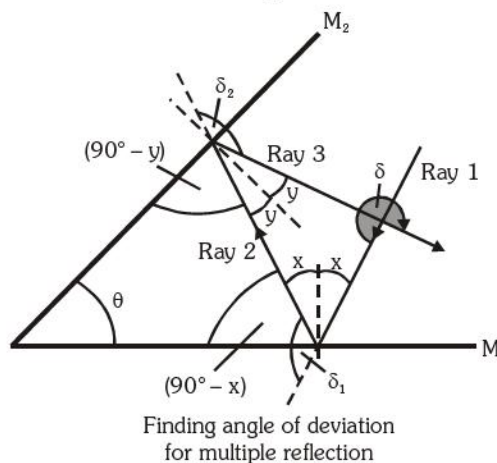
$$\text{or } \theta + 180^\circ - x - y = 180^\circ$$

$$\text{or } \theta - x - y = 0$$

$$\text{or } \theta = x + y \text{ ---- (2)}$$

From (1) & (2), we get,

$$\delta = 360^\circ - 2\theta$$



Example : Prove that for two plane mirrors which are placed perpendicular to each other, the emergent ray (final ray) is always parallel to incident ray whatever be the angle of incidence.

Solution :

Let us take two mirrors M_1 and M_2 placed perpendicular to each other. Let an incident ray (Ray 1) strikes the mirror M_1 at an angle of incidence x . By law of reflection,

angle of incidence = angle of reflection

$$\text{or } \angle ABE = \angle CBE = x$$

The ray reflected from mirror M_1 (Ray 2) strikes the mirror at an angle y . Again, by law of reflection,

angle of incidence = angle of reflection

$$\text{or } \angle BCD = \angle DCE = y$$

In $\triangle BCE$,

$$x + y + 90^\circ = 180^\circ$$

($\angle$ sum property of a $\triangle$)

$$\text{or } x + y = 90^\circ \text{ ---- (1)}$$

Now, from, $\angle ABC + \angle BCD = 2x + 2y = 2(x + y) = 2 \times 90^\circ$ [Using eq.(1)]

$$\text{or } \angle ABC + \angle BCD = 180^\circ$$

Since the pair of co-interior angles are supplementary, **$AB \parallel CD$**

That is, Ray 1 is parallel to Ray 3, but they are in opposite directions. Such rays are sometimes also called **antiparallel rays** (Here, angle of deviation is 180° as the Ray 1 and Ray 3 are in opposite directions).

Example : Two plane mirrors are inclined to each other such that a ray of light incident on the first mirror and parallel to the second mirror when reflected from the second mirror becomes parallel to the first mirror. Find the angle between the two mirrors.

Solution :

Given that Ray 1 $\parallel$ Mirror M_2 and Ray 2 $\parallel$ Mirror M_1

Let angle $\angle DCB = \theta$

Now, $\angle FDE = \angle DCB = \theta$ (corresponding $\angle$ s)

Now, $y + \angle FDE = 90^\circ$

$$\text{or } y + \theta = 90^\circ \text{ ---- (1)}$$

$$\text{Also, } \angle CDB + y = 90^\circ \text{ ---- (2)}$$

From (1) & (2), we get, $y + \theta = \angle CDB + y$

$$\text{or } \angle CDB = \theta \text{ ---- (3)}$$

$$\text{Similarly, } \angle DBC = \theta \text{ ---- (4)}$$

In $\triangle DBC$, $\angle DBC + \angle DCB + \angle CDB = 180^\circ$ ($\angle$ sum property of a $\triangle$)

$$\text{or } \theta + \theta + \theta = 180^\circ \quad \text{or} \quad 3\theta = 180^\circ \quad \text{or} \quad \theta = 60^\circ$$

● **Rotation of plane mirror :**

Let us take a plane mirror M and a given incident ray AO . Let ON be the normal at the point of incidence in this position of the mirror. Let the reflected ray be OB .

By law of reflection,

angle of incidence = angle of reflection,

$$\text{or } \angle AON = \angle BON = x \text{ (let) ---- (1)}$$

Let the mirror is now turned clockwise by an angle θ .

Thus, the normal will also rotate by the same angle θ .

Let the new normal be ON' . Thus, new angle of incidence will be (see fig.) $(x + \theta)$. Let the new reflected ray be OB' . Again, using law of reflection,

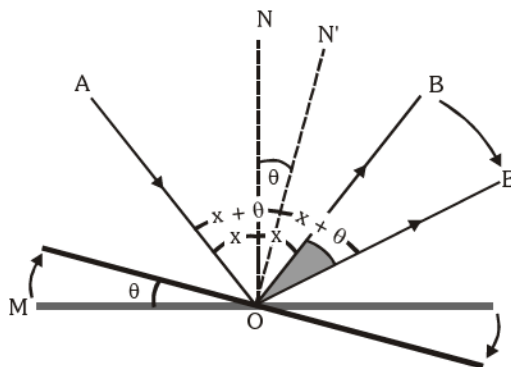
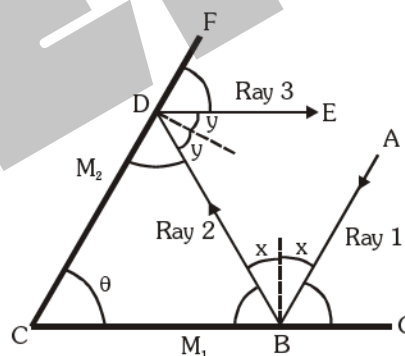
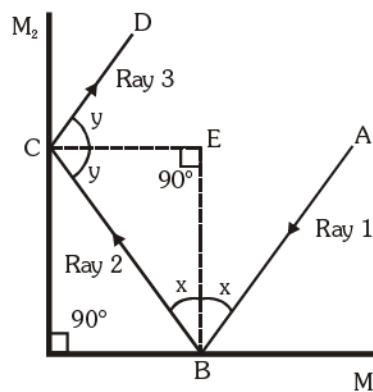
$$\angle AON' = \angle B'ON' = x + \theta \text{ ---- (2)}$$

$$\text{Now, } \angle NOB' = \angle NON' + \angle B'ON' = \theta + (x + \theta) = 2\theta + x \text{ ---- (3) [Using eq.(2)]}$$

$$\text{Also, } \angle NOB' = \angle BON + \angle BOB'$$

$$\text{or } 2\theta + x = x + \angle BOB' \text{ [Using eq.(1) \& eq.(3)]}$$

$$\text{or } \angle BOB' = 2\theta$$



- Thus, for a fixed incident light ray, if the mirror be rotated by an angle ' θ ', the reflected ray turns by an angle ' 2θ ' in the same direction. For example, if the mirror is rotated by an angle 15° clockwise, the reflected ray is rotated by an angle 30° clockwise.

● Rotation of incident ray

Let us take a plane mirror M and a given incident ray AO. Let ON be the normal at the point of incidence in this position of the mirror. Let the reflected ray be OB. By law of reflection,

$$\text{angle of incidence} = \text{angle of reflection},$$

$$\text{or } \angle AON = \angle BON = x \text{ (let)} \quad \text{---- (1)}$$

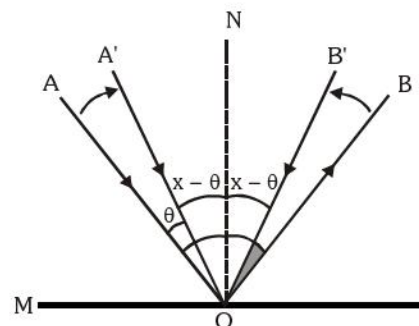
Now, the incident ray is turned by an angle θ in clockwise direction. Let the new incident ray be A'O. From fig., the new angle of reflection is, $\angle A'ON = x - \theta$. Again using, law of reflection,

$$\angle A'ON = \angle B'ON = x - \theta \quad \text{---- (2)}$$

Now, $\angle BON = \angle B'ON + \angle B'OB$

$$\text{or } x = x - \theta + \angle B'OB$$

$$\text{or } \angle B'OB = \theta$$

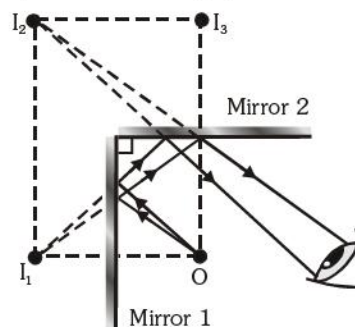
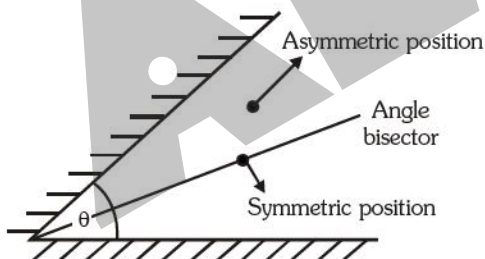


- Thus, for a fixed mirror, if the incident ray be rotated by an angle ' θ ', the reflected ray turns by an angle ' θ ' in opposite direction. For example, if the incident ray is rotated by an angle 30° clockwise, the reflected ray is rotated by angle 30° anticlockwise.

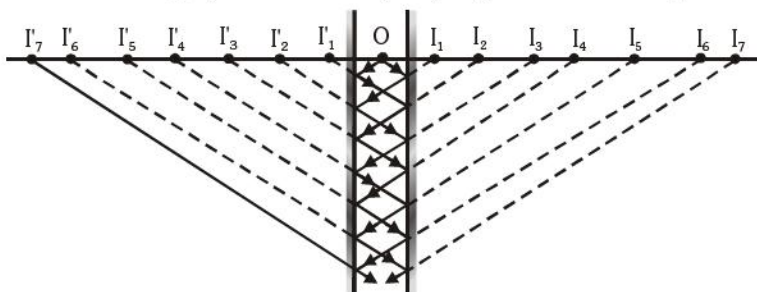
Multiple reflection of light by two plane mirrors inclined with each other :

$$\text{First find } m = \frac{360}{\theta}.$$

- If m is an even number, then for symmetric as well as asymmetric position, the number of images, $n = m - 1$
- If m is an odd number, then for symmetric position, $n = m - 1$ & for asymmetric position, $n = m$
- If m is a fraction and the object is symmetrically placed then, number of images, $n =$ nearest even integer.



- If $\theta = 90^\circ$, then, $n = 3$ for any position of the object placed between the two mirrors.
- If $\theta = 0^\circ$, the number of images, $n =$ infinite i.e., very large number of images are formed.



Images of an object formed by multiple reflection when placed between two parallel mirrors

Example : Find the number of images formed when the mirrors are placed at certain angle θ and the object is placed symmetrically between the mirrors : (1) $\theta = 60^\circ$ (2) $\theta = 72^\circ$ (3) $\theta = 50^\circ$.

Solution :

(1) Given, $\theta = 60^\circ$. First we will find, $m = \frac{360}{\theta} = \frac{360}{60} = 6$

Since, $m = 6$, an even number, no. of images is always $n = m - 1 = 6 - 1 = 5$

Thus, **5 images** are formed.

(2) Given, $\theta = 72^\circ$. First we will find, $m = \frac{360}{\theta} = \frac{360}{72} = 5$

Since, $m = 5$, an odd number, no. of images for a symmetrically placed object is given by,

$$n = m - 1 = 5 - 1 = 4$$

Thus, **4 images** are formed.

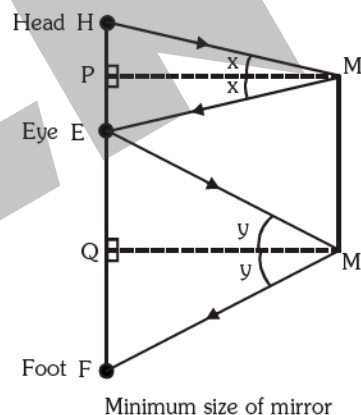
(3) Given, $\theta = 50^\circ$. First we will find, $m = \frac{360}{\theta} = \frac{360}{50} = 7.2$

Since, m is not an integer, it is a fraction, no. of images for a symmetrically placed object is the nearest even integer. The even integer nearest to the number 7.2 is 8. Thus, **8 images** are formed.

Some more points related to plane mirrors :

- (1) The minimum size of mirror required for a person to see his/her own complete (full height) image is 'half of the height of that person'. Let us understand mathematically :

To see his/her own full height image, the light rays from the head (highest point) and from the foot (lowest point) must reach his/her eyes. See fig. which shows this situation clearly. To find the minimum size of mirror, the rays from head should reflect from topmost point M_1 of the mirror and the rays from the foot should reflect from the lowermost point M_2 of the mirror. From fig.14,



Minimum size of mirror

$$\angle HM_1P = \angle EM_1P \text{ (Law of reflection)}$$

$$\angle M_1PH = \angle M_1PE = 90^\circ$$

$$PM_1 = PM_1 \text{ (common)}$$

$$\therefore \triangle HPM_1 \cong \triangle EPM_1 \text{ (ASA congruency)}$$

$$\therefore HP = PE \text{ ---- (1)}$$

$$\text{Similarly, } EQ = QF \text{ ---- (2)}$$

$$\text{Also, } PQ = M_1M_2 \text{ (Opposite sides of a rectangle)---- (3)}$$

$$\text{Now, } HF = HP + PE + EQ + QF = PE + PE + EQ + EQ \text{ [Using eq.(1) \& eq.(2)]}$$

$$\text{or } HF = 2PE + 2EQ = 2(PE + EQ) = 2PQ$$

$$\text{or } HF = 2M_1M_2 \text{ [Using eq.(3)]}$$

$$\text{or } M_1M_2 = \frac{1}{2}HF = \text{half of the height of the person}$$

- (2) If an object moves with speed 'u' towards a fixed mirror, the image also moves towards the mirror with speed 'u'. The speed of the image relative to the object is '2u'. Similarly, if the mirror moves with speed 'u' towards or away from a fixed object, the image appears to move towards or away from object with speed '2u'.
- (3) If the mirror moves away or towards an object by a distance 'd', then image moves away or towards the object by a distance '2d'.

Example : A man is standing at the middle of a cuboidal room having height H . Find the minimum size of mirror required to see the full image of the wall behind the man.

Solution : To see the full size image of any object, the rays from the topmost point and lowermost point must reach into our eyes (see fig.). Let the length of the room be L . The man is standing at the middle i.e., at a distance $L/2$ from the either walls. Let M_1M_2 be the mirror placed on front wall. In ΔAM_1P and ΔEM_1C , $\angle AM_1P = \angle EM_1C$ (law of reflection)
 $\angle APM_1 = \angle ECM_1 = 90^\circ$

By AA similarity, $\Delta AM_1P \sim \Delta EM_1C$

$$\therefore \frac{AP}{CE} = \frac{M_1P}{M_1C} = \frac{L}{(L/2)} = 2 \quad \text{or} \quad AP = 2CE \quad \text{---- (1)}$$

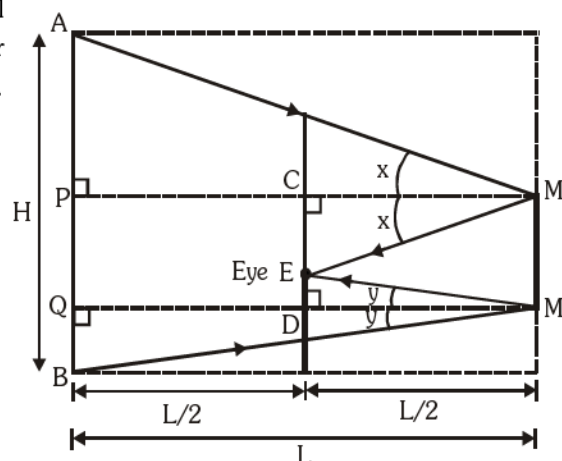
$$\text{Similarly, } BQ = 2ED \quad \text{---- (2)}$$

$$\text{Also, } PQ = CD = M_1M_2 \quad \text{---- (3)}$$

$$AB = AP + PQ + QB = 2CE + M_1M_2 + 2ED \quad [\text{Using eq.(1), eq.(2) \& eq.(3)}]$$

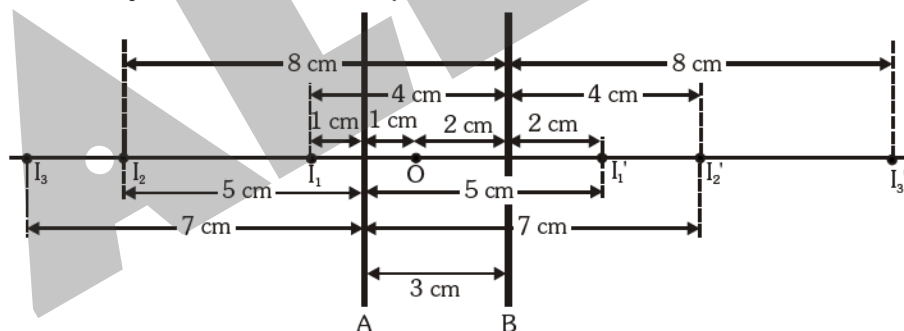
$$\text{or } AB = 2(CE + ED) + M_1M_2 = 2CD + M_1M_2 = 2M_1M_2 + M_1M_2 = 3M_1M_2$$

$$\text{or } M_1M_2 = \frac{1}{3} AB = \frac{H}{3} = \text{one-third of the height of the wall}$$



Example : An object is placed between two plane parallel mirrors A and B, which are 3 cm apart, so that it is 2 cm from the mirror B. Find the distance between third image behind A and third image behind B.

Solution : Let us make a figure (see fig.) and use the fact that in case of plane mirrors distance of the image from the mirror is equal to distance of the object from the mirror.



Here, I_1 is the image of O in mirror A and I_1' is the image of O in mirror B . I_1 serves as object for mirror B and forms image I_2' in mirror B . Similarly, I_1' serves as object for mirror A and forms image I_2 in mirror A , and so on. From fig. 16, distance between 3rd image on either side, $I_3I_3' = 7 \text{ cm} + 3 \text{ cm} + 8 \text{ cm} = \mathbf{18 \text{ cm}}$.

Note : You can use a direct formula to find the distance between n th image on either side = $2na$

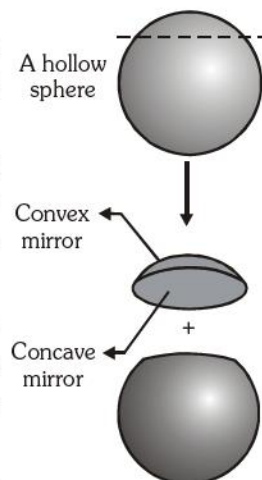
Where, a = distance between the two mirrors.

In this case, $n = 3$ and $a = 3 \text{ cm}$, thus, $I_3I_3' = 2na = 2 \times 3 \times 3 = \mathbf{18 \text{ cm}}$.

- If an object moves with speed ' u ' towards a fixed mirror, the image also moves towards the mirror with speed ' u '. The speed of the image relative to the object is ' $2u$ '. Similarly, if the mirror moves with speed ' u ' towards or away from a fixed object, the image appears to move towards or away from object with speed ' $2u$ '.
- If the mirror moves away or towards an object by a distance ' d ', then image moves away or towards the object by a distance ' $2d$ '.

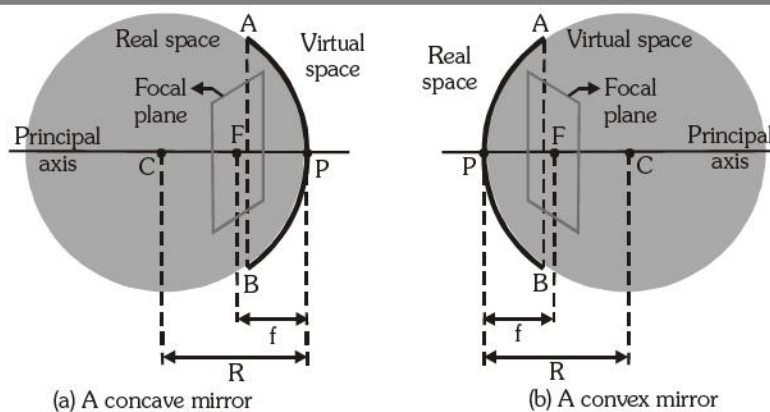
■ Reflection from spherical mirrors

- A spherical mirror, as the name suggests, has the shape of a section of a hollow sphere. A **spherical mirror** is a mirror whose reflecting surface is made by the part of a hollow sphere.
- Spherical mirror can be thought of as a portion of a sphere that was sliced away and then silvered on one of the sides to form a reflecting surface. Concave mirror is silvered on the inside of the sphere and convex mirror is silvered on the outside of the sphere.
- Suppose a hollow sphere has a polished mirror surface on the inside as well outside. By removing a section of the sphere a double-sided spherical mirror is obtained with a concave reflecting surface on one side and a convex reflecting surface on the other (see fig.).
 - ▶ The concave reflecting surface is curved inwards. The convex reflecting surface is curved outwards.
- **Concave mirror** : A spherical mirror in which the reflection of light takes place at bent-in surface is called 'concave mirror'.
 - ▶ Concave mirror is also called '**converging mirror**'. This is because the parallel beam of light after reflection, converge at a single point.
- **Convex mirror** : A spherical mirror in which the reflection of light takes place at bulging-out surface is called 'convex mirror'.
 - ▶ Convex mirror is also called '**diverging mirror**'. This is because the parallel beam of light after reflection appears to diverge from a single point.



■ Basic terms related to spherical mirrors

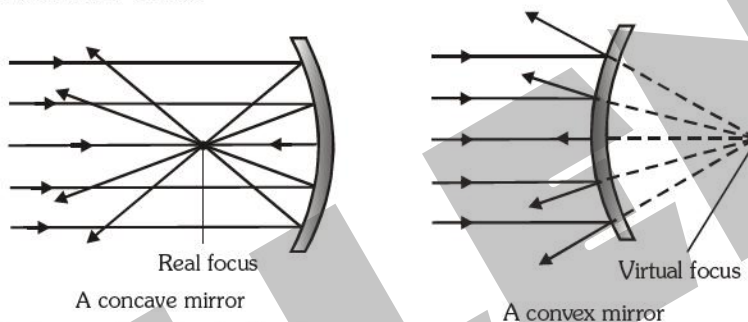
- **Centre of curvature** : The point in space that represents the centre of the hollow sphere from which the spherical mirror was cut is called 'centre of curvature'.
 - ▶ The centre of the hollow sphere from which the spherical mirror is formed is called '**centre of curvature**'.
- **Pole (or vertex)** : The middle point on the surface of a spherical mirror is called 'pole'.
 - ▶ The geometric centre of the curved mirror surface is called '**pole**'.
- **Radius of curvature** : The radius of hollow sphere from which the mirror is formed is called 'radius of curvature'.
 - ▶ The distance between the centre of curvature and the pole of a spherical mirror is called '**radius of curvature**'.
- **Principal axis** : A line passing through the pole and the centre of curvature of the spherical mirror is called 'principal axis'.
 - ▶ An imaginary line drawn through the pole, perpendicular to the surface of the spherical mirror at the pole is called '**principal axis**'.
- **Principal focus** : The point on the principal axis where all rays parallel to principal axis either converge or appear to diverge after reflection is called 'principal focus'.
- **Focal length** : The distance between the focus and pole of a spherical mirror is called 'focal length'.
- **Focal plane** : A plane passing through focus and perpendicular to principal axis is called 'focal plane'.
- **Aperture** : The diameter of the circular cross-section of the spherical mirror is called 'aperture' (AB). It represents the size of the mirror. More the aperture of a spherical, more will be its size. Thus, it will collect more light forming brighter images after reflection.



C : Center of curvature ; F : Principal focus ; P : Pole or vertex
f : focal length ; R : Radius of curvature ; AB : Aperture

Basic geometry and terms of concave and convex mirrors

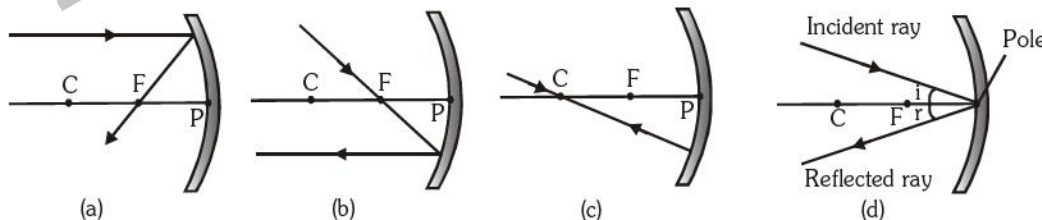
- For concave mirrors, centre of curvature and principal focus are real. For convex mirrors, centre of curvature and principal focus are virtual.



Rules to obtain images in spherical mirrors

Concave mirrors

- The ray parallel to the principal axis, after reflection, passes through the principal focus F of a concave mirror.
- A ray passing through the principal focus in a concave mirror is reflected parallel to the principal axis.
- A ray passing through the centre of curvature in a concave mirror is reflected back along its own path.
- When a ray is incident obliquely to the principal axis, towards the pole of the mirror, on a concave mirror, it is reflected obliquely such that the incident ray and the reflected ray make equal angles with the principal axis. This is because the principal axis acts as normal at the pole. The incident and reflected rays follow the laws of reflection at the point of incidence (pole), making equal angles with the principal axis.

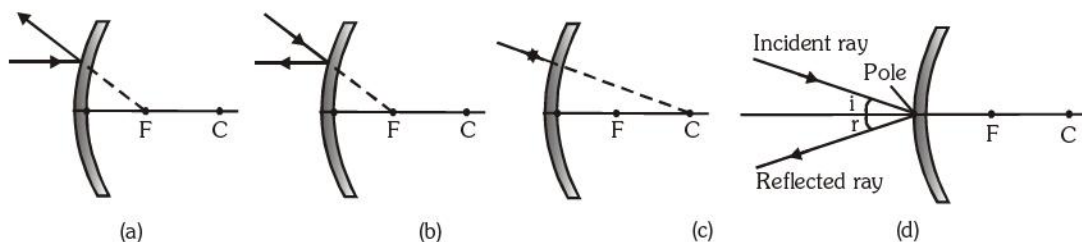


Rules to obtain images for a concave mirror

Convex mirrors

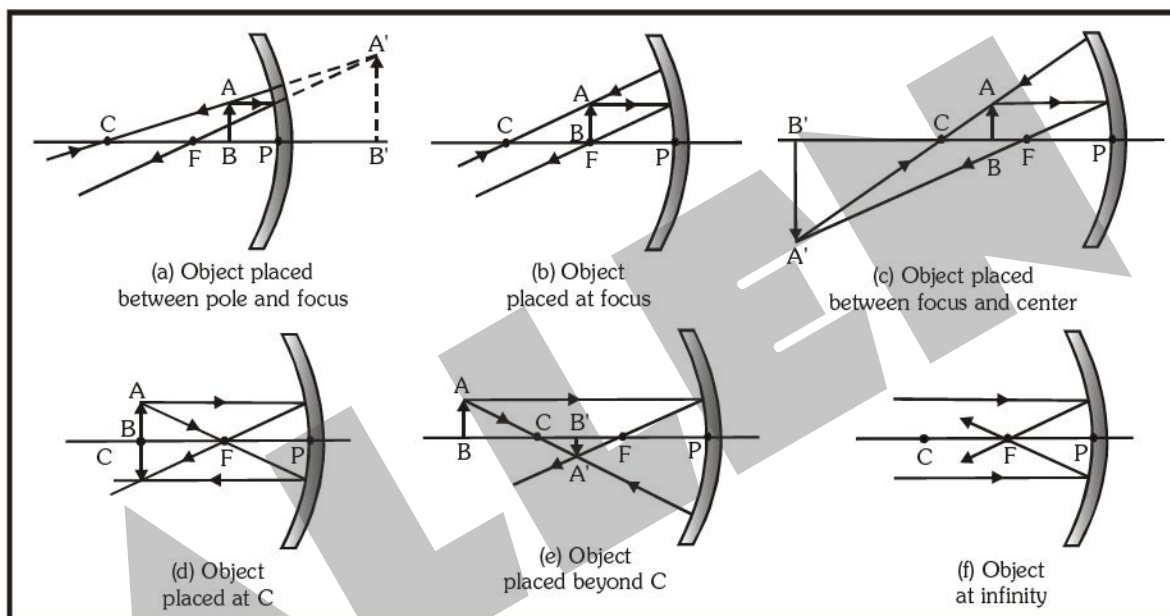
- The ray parallel to the principal axis, after reflection, appears to diverge (pass) through the principal focus of a convex mirror.
- A ray which is directed towards the principal focus in a convex mirror is reflected parallel to the principal axis.
- A ray directed towards the centre of curvature in a convex mirror, is reflected back along its own path.

- When a ray is incident obliquely to the principal axis, towards the pole of the mirror, on a convex mirror, it is reflected obliquely such that the incident ray and the reflected ray make equal angles with the principal axis. This is because the principal axis acts as normal at the pole. The incident and reflected rays follow the laws of reflection at the point of incidence (pole), making equal angles with the principal axis.



Rules to obtain images for a convex mirror

■ Image formation by a concave mirror



- For a spherical mirror, the line joining the centre of curvature and any point on the mirror represents the radius of curvature and it acts as a normal at that point.

Image formation by a concave mirror

Position of the object	Position of the image	Size of the image	Nature of the image
Between P and F	Behind the mirror	Enlarged	Virtual and erect
At F	At infinity	Highly enlarged	Real and inverted
Between C and F	Beyond C	Enlarged	Real and inverted
At C	At C	Same size	Real and inverted
Beyond C	Between F and C	Diminished	Real and inverted
At infinity	At the focus F	Highly diminished, point-sized	Real and inverted

● Uses of concave mirrors

- Concave mirrors are used as shaving mirrors to see a larger image of the face.
- Concave mirrors are used as reflectors in car head lights, search lights, hand torches, table lamps, etc. to get powerful parallel beams of light.
- Concave mirrors are used in solar power plants to produce electricity.
- Concave mirrors are used by doctors to concentrate light on body parts like ears and eyes.
- Concave mirrors are also used by dentists to see large images of the teeth of patients.

- **Example :** Suppose that lower half of concave mirror's reflecting surface is covered with an opaque (non-reflecting) material. What effect will this have on the image of an object placed in front of the mirror ?

Solution : If lower half of concave mirror is obstructed, full image will be formed but with reduced brightness. This is because every part of mirror forms complete image. On obstructing the lower half, half of the light rays are obstructed, forming less bright image. In this case, the intensity (brightness) of image will be half the initial value.

- **Image formation by a convex mirror**

The image formed by a convex mirror is always behind the mirror that is, it is always virtual and erect. Also, the size of image is always diminished, that is, its size is always smaller than that of the object.

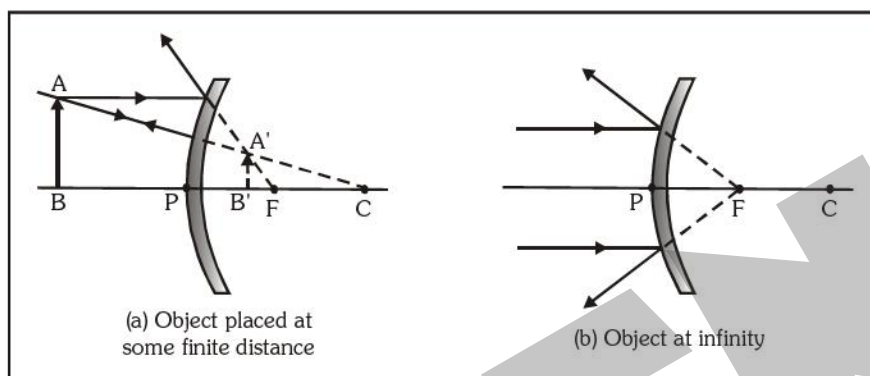


Image formation by a convex mirror

Position of the object	Position of the image	Size of the image	Nature of the image
Between infinity and the pole P	Between P and F, behind the mirror	Diminished	Virtual and erect
At infinity	At the focus F, behind the mirror	Highly diminished, point-sized	Virtual and erect

- **Uses of convex mirrors :**

Convex mirrors are commonly used as rear-view (wing) mirrors in vehicles. These mirrors are fitted on the sides of the vehicle, enabling the driver to see traffic behind him/her to facilitate safe driving.

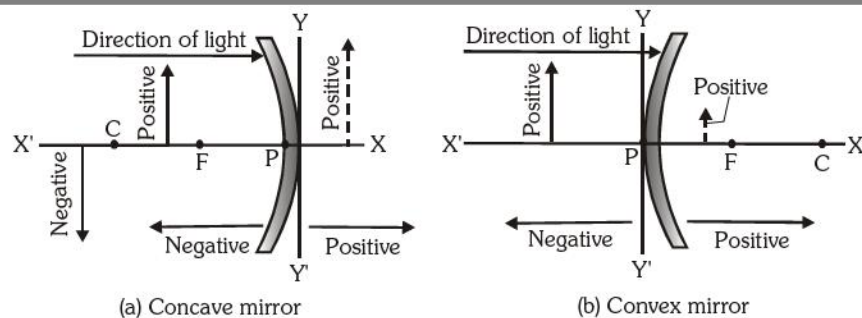
- Convex mirrors are preferred as rear view mirrors because they always give an erect, though diminished, image. Also, they have a wider field of view as they are curved outwards. Thus, convex mirrors enable the driver to view much larger area than would be possible with a plane mirror.

- **Sign convention for reflection by spherical mirrors**

While dealing with the reflection of light by spherical mirrors, we follow a set of sign conventions called the **new Cartesian sign convention**. In this convention, the pole (P) of the mirror is taken as the origin. The principal axis of the mirror is taken as the x-axis (XX') of the coordinate system.

The conventions are as follows :

- The object is always placed to the left of the mirror. This implies that the light from the object falls on the mirror from the left-hand side.
- All distances parallel to the principal axis are measured from the pole of the mirror. All the distances along XX' axis are measured from P.
- All the distances measured to the right of the origin (along + x axis) are taken as **positive** while those measured to the left of the origin (along - x axis) are taken as **negative**.
- Distances measured perpendicular to and above the principal axis (along + y-axis) are taken as **positive**. Distances measured perpendicular to and below the principal axis (along -y-axis) are taken as **negative**.



The new Cartesian sign convention for spherical mirrors

■ Formulae related to spherical mirrors

- **Relationship between radius of curvature and focus** : The focal length of a spherical mirror is equal to half of its radius of curvature i.e.,

$$f = \frac{R}{2}$$

- **Mirror formula** : In a spherical mirror, the distance of the object from its pole is called the **object distance (u)**. The distance of the image from the pole of the mirror is called the **image distance (v)**. The relationship between object distance (u), the image distance (v) and the focal length (f) is given by mirror formula which is as given below,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

- **Magnification (m)** : The ratio of height of image (h_2) to the height of object (h_1) is called 'magnification' or 'linear magnification'.

$$m = \frac{h_2}{h_1}$$

- ▶ The magnification m is also related to the object distance (u) and image distance (v). It can be expressed as:

$$m = \frac{h_2}{h_1} = \frac{-v}{u}$$

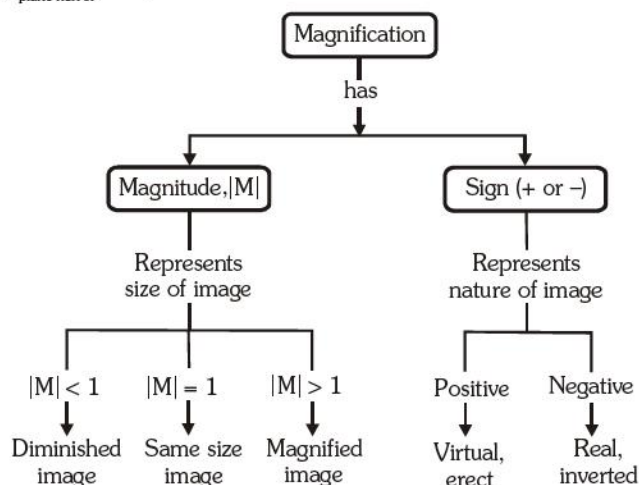
Also, magnification can be further expressed as,

$$m = \frac{f}{f-u} = \frac{f-v}{f}$$

Example : A spherical mirror produces a magnification of +1.5. Explain the nature and size of the image formed by it. Which type of spherical mirror is this?

Explanation : Since, the sign of magnification is positive, this means the image is virtual and erect. Now, $|M| = 1.5$, which is greater than one, this means the image is magnified. The spherical mirror in this case is a concave mirror (a converging mirror) as it produces a virtual, erect and magnified image.

- A plane mirror forms a virtual, erect and same size image that of the object thus, the magnification of a plane mirror is +1. Radius of curvature of a plane mirror is infinite. Thus, focal length ($f = R/2$) of a plane mirror is also infinite i.e., $f_{\text{plane mirror}} = \infty$.



■ Some important points related to spherical mirrors

● Concave mirror

- ▶ Object distance, u = always negative.
- ▶ Image distance, v = positive, when object is placed between P & F (virtual and erect image).
 v = negative, all other possible cases (real and inverted image).
- ▶ f = negative, R = negative.

● Convex mirror

- ▶ Object distance, u = always negative.
- ▶ Image distance, v = always positive (virtual and erect).
- ▶ f = positive, R = positive.
- ▶ Image is always diminished.

- **Example :** A convex mirror used for rear-view on an automobile has a radius of curvature of 3.00 m. If a bus is located at 5.00 m from this mirror, find the position, nature and size of the image.

Solution : Given, radius of curvature, $R = +3$ m; object distance, $u = -5$ m ; image distance, $v = ?$; magnification, $m = ?$

Focal length, $f = R/2 = +3/2$ m

Mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad \frac{1}{v} + \frac{1}{(-5)} = \frac{1}{(+3/2)} \quad \text{or} \quad \frac{1}{v} - \frac{1}{5} = +\frac{2}{3} \quad \text{or} \quad \frac{1}{v} = \frac{1}{5} + \frac{2}{3} = \frac{3+10}{15} = \frac{13}{15}$$

or $v = \frac{15}{13} = +1.15$ m Thus, the image is 1.15 m at the back of the mirror.

$$\text{Now, magnification, } m = \frac{h_2}{h_1} = -\frac{v}{u} = -\frac{(+15/13)}{(-5)} = +\frac{3}{13} = +0.23$$

The image is virtual, erect and smaller in size by a factor of 0.23.

- **Example :** An object, 4.0 cm in size, is placed at 25.0 cm in front of a concave mirror of focal length 15.0 cm. At what distance from the mirror should a screen be placed in order to obtain a sharp image ? Find the nature and the size of the image.

Solution : Given, object size, $h_1 = +4$ cm ; object distance, $u = -25$ cm ; focal length, $f = -15$ cm ; image distance, $v = ?$; image size, $h_2 = ?$

Mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad \frac{1}{v} + \frac{1}{(-25)} = \frac{1}{-15} \quad \text{or} \quad \frac{1}{v} - \frac{1}{25} = -\frac{1}{15} \quad \text{or} \quad \frac{1}{v} = \frac{1}{25} - \frac{1}{15} = \frac{3-5}{75} = \frac{-2}{75}$$

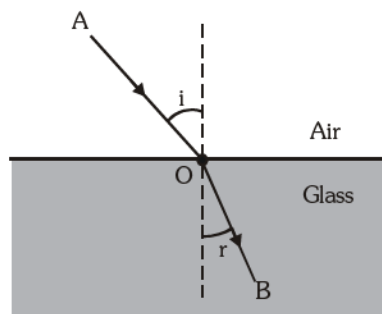
or $v = \frac{-75}{2} = -37.5$ cm The screen should be placed at 37.5 cm from the mirror.

$$\text{Now, magnification, } m = \frac{h_2}{h_1} = -\frac{v}{u} \quad \text{or} \quad h_2 = -\frac{v}{u} \times h_1 = -\frac{(-75/2)}{(-25)} \times (+4) = -\frac{3}{2} \times 4 = -6 \text{ cm}$$

■ Refraction of light

When a ray of light travels from one transparent medium to another, it bends while crossing the surface separating the two media.

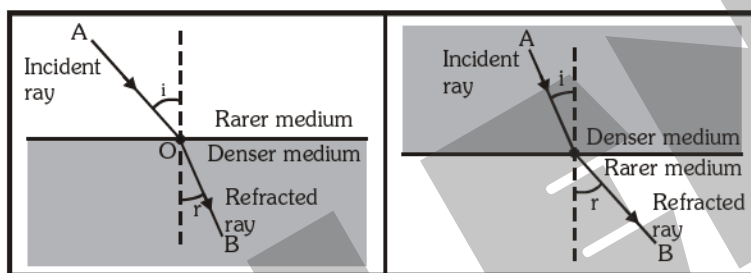
- The phenomenon of change in path of light when it passes from one medium to another is called '**refraction**'.
- **Incident ray :** The ray of light falling on the surface of a transparent medium is called '**Incident ray**'.
- **Refracted ray :** The ray of light which bends after passing through a transparent medium is called '**refracted ray**'.
- **Angle of incidence :** Angle made by the incident ray with the normal at point of incidence is called '**angle of incidence**'.
- **Angle of refraction :** Angle made by the refracted ray with the normal at point of incidence is called '**angle of refraction**'.
- **Cause of refraction :** The bending of light takes place when it passes from one medium to another because speed of light changes from one medium to another. Speed of light is different in different media.



- **Optically rarer medium** : A transparent medium in which the speed of light is more is called 'optically rarer medium' (or simply 'rarer medium').
- **Optically denser medium** : A transparent medium in which the speed of light is less is called **optically denser medium** (or simply 'denser medium').
 - ▶ Speed of light is maximum in vacuum, it is about 3×10^8 m/s. Speed of light in water is approximately 2.25×10^8 m/s.
 - ▶ The frequency of light remains constant during the refraction of light. The velocity and wavelength of light changes during refraction of light. Also, amplitude and intensity of light decreases during refraction of light due to absorption of small amount of light at the surface. As $v = v\lambda$ thus, $v \propto \lambda$. This means velocity of light is directly proportional to wavelength. If velocity decreases, wavelength also decreases and vice-versa.

$$\frac{v_1}{\lambda_1} = \frac{v_2}{\lambda_2} \quad \text{or} \quad \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

- **Refraction of light from rarer to denser medium** : When ray of light passes from rarer medium to denser medium, the refracted ray bends towards the normal at the point of incidence.
- **Refraction of light from denser to rarer medium** : When ray of light passes from denser medium to rarer medium, the refracted ray bends away from the normal at the point of incidence.



S.No.	Refraction through	
	Rarer to denser	Denser to rarer
1.	Refracted ray bends towards the normal	Refracted ray bends away from the normal
2.	Frequency unchanged	Frequency unchanged
3.	Velocity decreases	Velocity increases
4.	Wavelength decreases	Wavelength increases
5.	Angle of incidence is greater than angle of refraction $\angle i > \angle r$	Angle of incidence is less than angle of refraction $\angle i < \angle r$

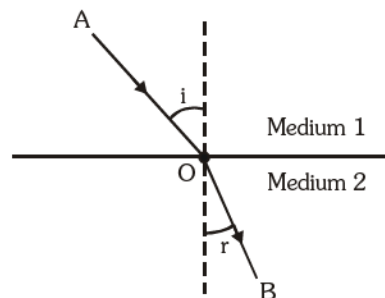
■ Laws of Refraction

- The incident ray, the normal to the refracting surface at the point of incidence and the refracted ray, all lie in the same plane
- The ratio of sine of angle of incidence to the sine of angle of refraction is constant for two given media. The constant is denoted by n_{21} and it is called '**refractive index of medium 2 with respect to medium 1**' (or simply '**relative refractive index**').

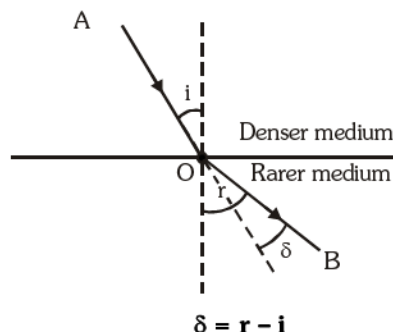
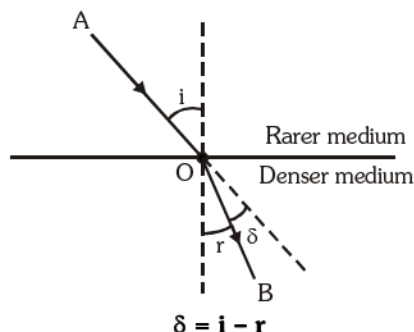
$$\frac{\sin i}{\sin r} = n_{21} \quad \text{The is called 'Snell's Law'}$$

- **Refractive index (n_{21}) depends on :**

- ▶ Nature of pair of media 1 and 2.
- ▶ Wavelength of the incident light ray. (Higher the wavelength, smaller will be the refractive index and vice-versa, valid in case of a dispersive medium). e.g. Refractive index is greater for violet light (shorter wavelength) and smaller for red light (longer wavelength).



- **Angle of deviation (δ)** : The angle through which the incident ray of light is deviated from its original path when it is refracted while passing from one transparent medium to another is called '**angle of deviation**'.
- ▶ The angle between incident ray and refracted ray is called '**angle of deviation**'.



- **Relative refractive index of a medium (n_{21})** :

The relative refractive index of a medium 2 with respect to medium 1 is the ratio of speed of light in medium 1 to the speed of light in medium 2.

$$n_{21} = \frac{\text{speed of light in medium 1}}{\text{speed of light in medium 2}} \quad \text{or} \quad \boxed{n_{21} = \frac{v_1}{v_2}}$$

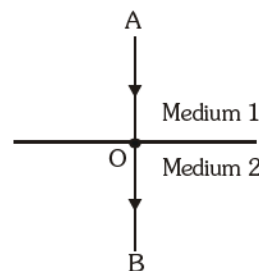
- **Absolute refractive index of a medium** : The ratio of speed of light in vacuum to the speed of light in a medium is called **absolute refractive index of medium**.

$$n = \frac{\text{speed of light in vacuum}}{\text{speed of light in medium}} \quad \text{or} \quad \boxed{n = \frac{c}{v}}$$

Where, c = speed of light in vacuum ; v = speed of light in medium

- Also, $v = \frac{c}{n}$ or $v \propto \frac{1}{n}$.

- ▶ Greater the value of ' n ', lesser will be the speed of light, medium will optically denser. Lesser the value of ' n ', greater will be the speed of light, medium will optically rarer.
- ▶ If the ray of light enters from medium 1 into medium 2 perpendicular to the surface of medium 2, the ray passes undeviated. That is, no bending of light occurs. In such a case, angle of incidence is zero, $\angle i = 0^\circ$ and $\angle r = 0^\circ$.

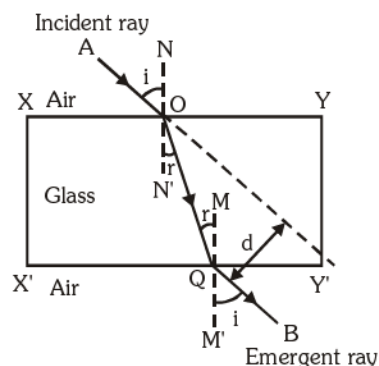


- $\boxed{n_{21} = \frac{1}{n_{12}}}$ Also, $\boxed{n_{21} = \frac{n_2}{n_1}}$

■ Refraction through a glass slab :

Let us consider a glass slab bounded by parallel faces XY and X'Y'. A ray of light travelling in air along the path AO gets incident on the face XY. On refraction into glass it bends towards the normal NN' and travels along the path OQ. The refracted ray OQ strikes the surface X'Y'. At X'Y', again refraction takes place, the ray of light bends away from normal MM'. This ray is called '**emergent ray**' and it is parallel to the incident ray. But, there is a shift from initial path (AO) in the emergent ray (QB), called '**lateral shift**'.

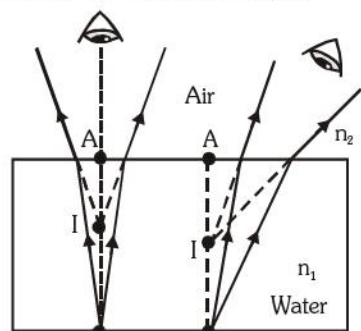
- Here, incident ray is parallel to the emergent ray but the incident ray is laterally displaced.



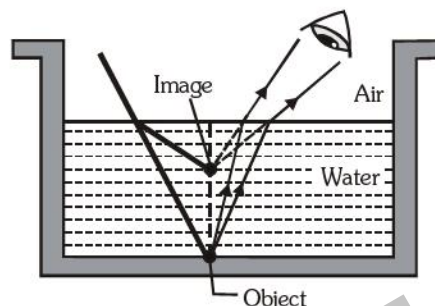
- **Apparent Depth :** When an object lying inside an optically denser medium is seen from a rarer medium, its depth appears to be less than its real depth. This depth is called '**apparent depth**' (see fig.).

Apparent depth = AI ; real (actual) depth = OA.

- A straight stick when immersed partly in water and held inclined to the surface, it appears to be bent at the point where it enters the water. This apparent bending of stick is also due to the refraction of light when it passes from water into air (see fig.).



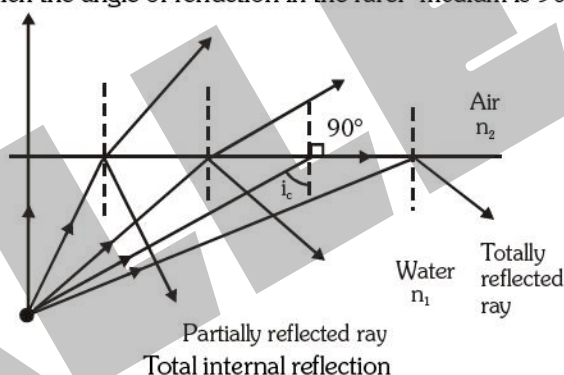
Apparent depth of an object is less than the actual depth when seen from air



A stick appears bent when placed in water due to refraction of light

- **Total Internal Reflection :** The phenomenon of reflection of light that takes place when a ray of light travelling in a denser medium gets incident at the interface of the two media at an angle greater than the critical angle for the given pair of media is called '**total internal reflection**'.

- **Critical angle :** The critical angle for a given pair of media is defined as 'the angle of incidence in the denser medium, for which the angle of refraction in the rarer medium is 90° '.

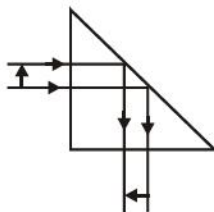


$$n_{12} = \frac{1}{\sin i_c} = \frac{n_1}{n_2}$$

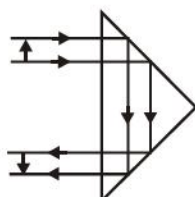
$$\text{or } \sin i_c = \frac{1}{n_{12}} = \frac{n_2}{n_1}$$

- **Some phenomena related to total internal reflection**

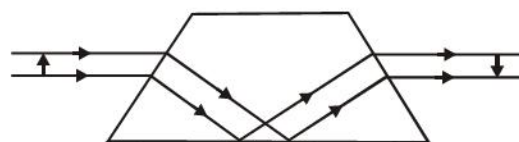
- ▶ The sparkling brilliance of diamonds is due to total internal reflection. The critical angle of diamond is very small (24.4°), so that once the light enters into the diamond, multiple internal reflections take place. This causes sparkling brilliance in diamond. By cutting the diamond suitably, multiple internal reflections can be made to occur.
- ▶ **Prism :** Prisms utilise the phenomenon of total internal reflection to bend light by 90° , or by 180° , or to invert images without changing their size. To bend light by 90° or 180° , the critical angle (i_c) for the material of prism should be less than 45° . e.g. dense flint glass ($i_c = 37.31^\circ$), crown glass ($i_c = 41.14^\circ$).



(a) Bending by 90°

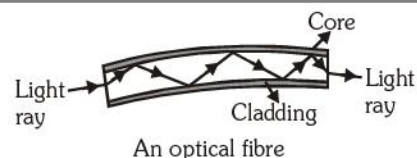


(b) Bending by 180°



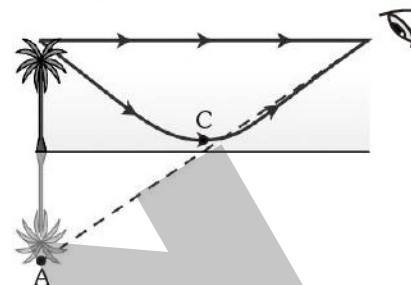
(c) Inverting an image

- **Optical fibres** : Total internal reflection is the basic principle used in optical fibres. The angle of incidence is always larger than the critical angle of core material with respect to cladding (outer part). A bundle of optical fibres can be put to several uses. It is called '**light pipe**'. A light pipe is used in medical and optical examination. e.g. In endoscopy, a patient's stomach can be viewed by inserting one end of light pipe into the stomach. The light is illuminated inside the stomach, the reflected light from inside travels through another set of fibres and viewer gets image at other end which can be magnified and seen on the screen of a computer monitor.



Optical fibres are used for transmitting and receiving electric signals which are converted to light by suitable transducers. The main requirement for transmission through optical fibres is very little absorption as it travels for long distances inside the optical fibre. In silica glass, it is possible to transmit more than 95 % of light over a fibre length of 1 km.

- **Mirage** : On still summer days, the air near the ground becomes hotter than air further up into the atmosphere. The refractive index of air increases with the increase in its density. Hotter air is less dense and thus has smaller refractive index than the cooler air. So, light from a tall object (a tree or a building) passes through a medium whose refractive index decreases towards the ground. Thus, a ray of light from such an object gets bent more and more away from the normal till it reaches a point (C) from where it gets totally internally reflected. Then, the light ray starts travelling up into the air regions of increasing refractive index. So, it bends more and more towards the normal and it appears to come from a point (A) in the ground. The observer views an inverted image of the object and naturally assumes that it is reflected from the surface of a water source like pond, lake, etc. But, such water source does not exist.

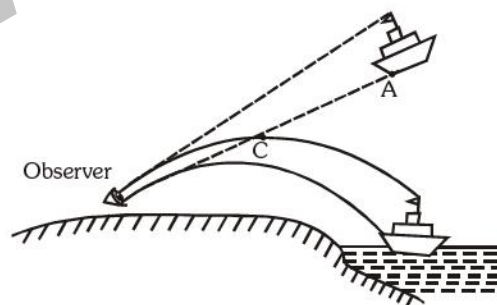


Mirage : an optical illusion

The inverted images of distant high objects seen on the summer days causing an optical illusion is called '**mirage**'. Mirages are specially a common phenomenon in '**hot deserts**'.

Mirage can be seen more frequently on a small scale as a wavy water surface over coal tarred roads or smooth concrete highways on hot summer days. On reaching closer, such water waves disappear as it is an optical illusion (called mirage).

- **Looming** : Looming is a particular type of mirage observed in very cold regions specially arctic region. When warmer air lies above the cooler air. The refractive index of air increases with the increase in density. Thus, in this particular case the refractive index of air increases on moving downwards as the cooler air which is more dense is present near the ground. A ray of light from an object like ship present in cold water gets bent gradually upward more and more away from the normal, it reaches a point (C) from where it gets totally internally reflected. Then, the light ray starts travelling down into the air regions of increasing refractive index. So, it bends gradually more and more towards the normal and it appears to come from a point (A) quite high into the atmosphere. The observer views an erect, virtual image of such an object at a height above the object. Thus, the object appears to be hanging in mid air, above the object itself.



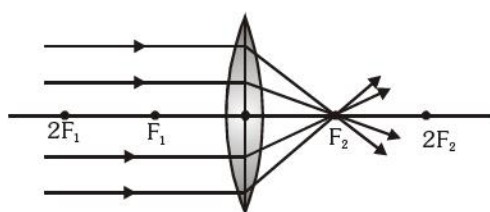
Looming : an optical illusion

■ **Lenses** : A lens is a piece of transparent glass bound by spherical surfaces.

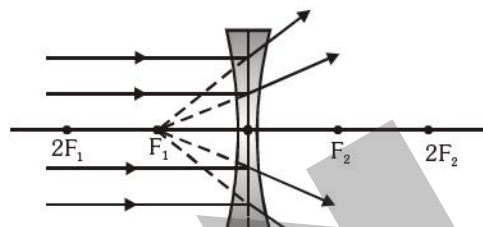
- **Convex Lens** : A convex lens is a piece of a transparent glass bound by two bulging out surfaces. It is thick at the middle and thinner at the edges. It is also called '**double convex lens**' or '**biconvex lens**'.
 - A convex lens is a '**converging lens**' because the light rays after refraction through it converge to meet at a single point.
- **Concave lens** : A concave lens is a piece of transparent glass bound by two bent-in (or bulging-in) surfaces. This is thin in the middle but thicker at the edges. It is also called '**Double concave**' or '**Biconcave lens**'.
 - A concave lens is a '**diverging lens**' because the parallel beam of light rays after refraction through it appear to diverge from a single point.

■ Basic terms used in lenses

- **Optical Centre** : The middle point of a spherical lens is called its 'optical centre'.
- **Principal Axis** : The line passing through the optical centre of the lens which is perpendicular to both the faces of the lens is called its 'principal axis'.
- **Principal Focus** : The point on the principal axis of a lens where all the rays parallel to principal axis converge or appear to diverge after refraction is called 'principal focus'.
- **Focal length** : The distance between the focus and the optical centre of a lens is called its 'focal length'.
 - ▶ A convex / concave lens has two principal foci F_1 and F_2 . F_1 is towards left of the lens and from this region light rays are incident on the lens. F_2 is towards right of the lens and in this region light rays are emergent after refraction.



A convex lens

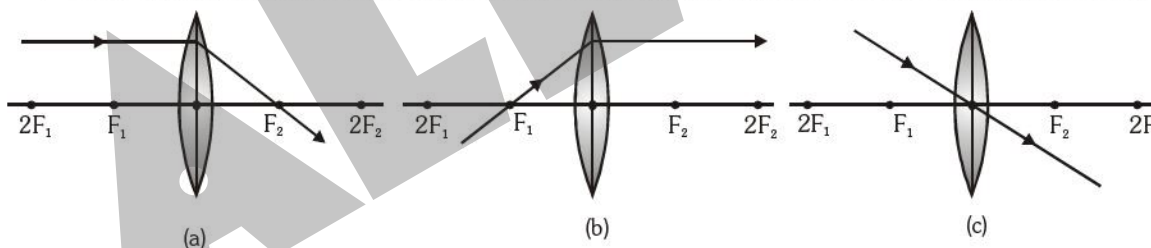


A concave lens

■ Rules to obtain images in spherical lenses

● Convex lens

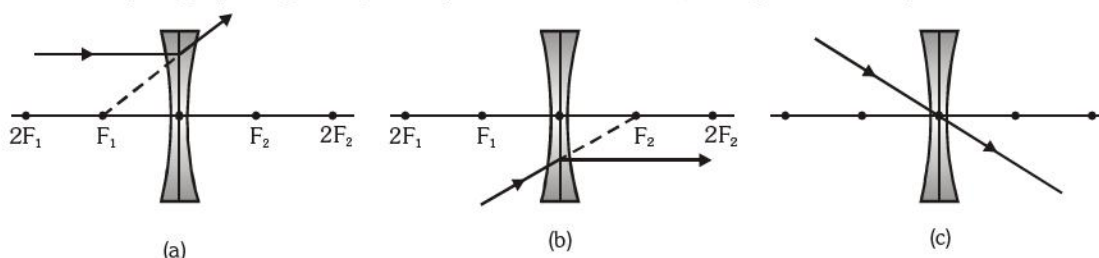
- ▶ A ray from the object parallel to the principal axis after refraction passes through the second principal focus F_2 .
- ▶ A ray of light passing through the first principal focus in a convex lens emerges parallel to the principal axis after refraction.
- ▶ A ray of light passing through the optical centre of the lens, emerges without any deviation after refraction.



Rules to obtain images for a convex lens

● Concave lens

- ▶ A ray from the object parallel to the principal axis after refraction appear to diverge from the first principal focus F_1 in a concave lens.
- ▶ A ray of light passing through the appearing to meet at second principal focus F_2 in a concave lens emerges parallel to the principal axis after refraction.
- ▶ A ray of light passing through the optical centre of the lens, emerges without any deviation after refraction.



■ Image formation by a convex lens

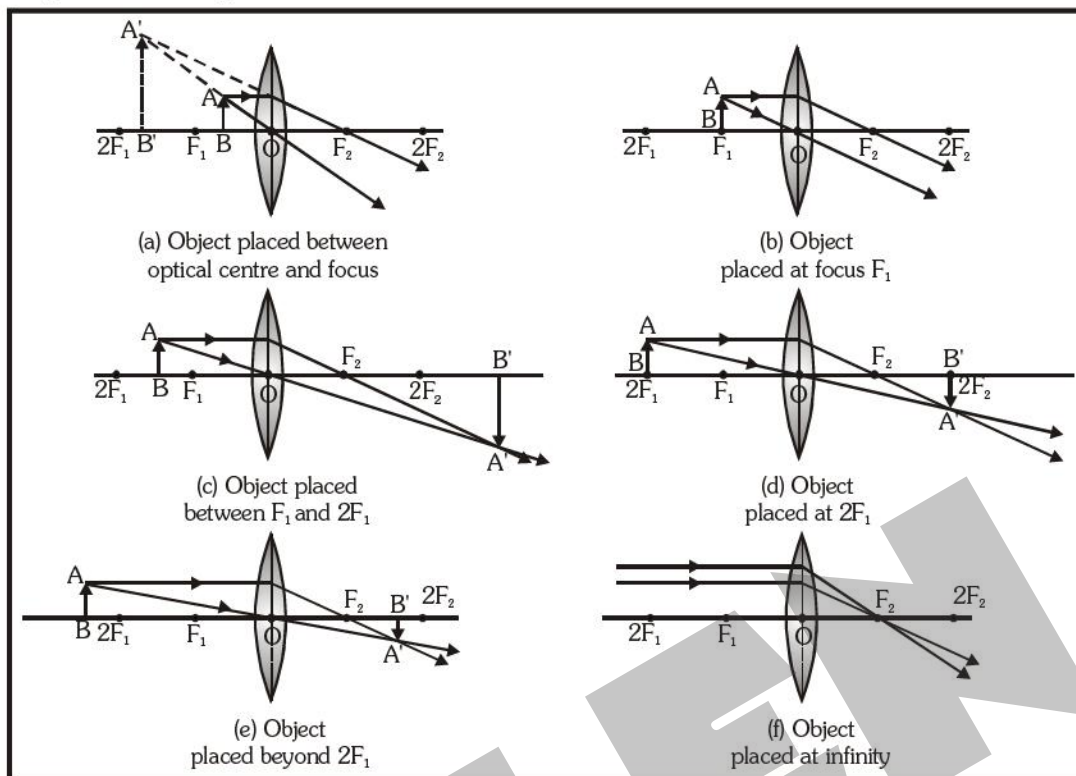


Image formation by a convex lens

Position of the object	Position of the image	Size of the image	Nature of the image
Between O & F_1	On the same side of the lens as the object	Enlarged	Virtual and erect
At F_1	At infinity	Highly enlarged	Real and inverted
Between F_1 & $2F_1$	Beyond $2F_2$	Enlarged	Real and inverted
At $2F_1$	At $2F_2$	Same size	Real and inverted
Beyond $2F_1$	Between F_2 & $2F_2$	Diminished	Real and inverted
At infinity	At focus F_2	Highly diminished, point-sized	Real and inverted

- A convex lens is used as a magnifier in simple microscope, compound microscope, telescope, etc.
- A convex lens is also used to correct eye defect 'hypermetropia' or 'long sightedness'.

■ Image formation by a concave lens

The image formed by a concave lens is always on the same side as the object and it is always virtual and erect. Also, the size of image is always diminished, that is, its size is always smaller than that of the object.

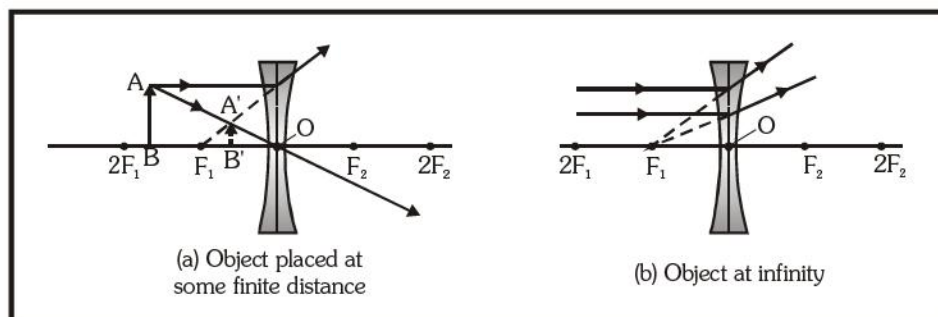


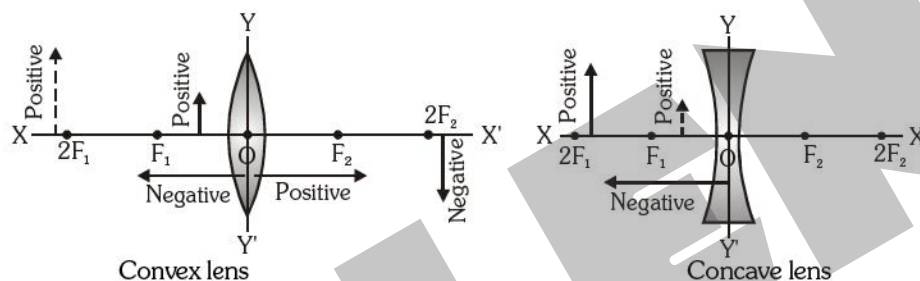
Image formation by a concave lens

Position of the object	Position of the image	Size of the image	Nature of the image
Between infinity & optical centre O	Between O & F_1 , i.e., same side as the object	Diminished	Virtual and erect
At infinity	At the focus F_1	Highly diminished, point-sized	Virtual and erect

- A concave lens is used to correct the eye defect 'myopia' or 'short sightedness'.

■ Sign conventions for lens

- Optical centre 'O' is taken as origin, all the distances along XX' axis are measured from 'O'
- Distance along direction of light are considered 'positive'.
- Distance measured opposite to the direction of light are considered 'negative'.
- Along YY', the height of object is always taken above principal axis and it is considered 'positive'. For image, if image is virtual and erect (above principal axis), the height is positive. If image is real and inverted (below principal axis), the height is 'negative'.



The new Cartesian sign convention for lenses

● Lens formula :

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

● Magnification :

$$m = \frac{h_2}{h_1} = \frac{v}{u} = \frac{f}{f+u} = \frac{f-v}{f}$$

- For **convex lens**, 'm' can be +ve as well as -ve. Also, |m| can be >1 or <1 or 1. For **concave lens**, 'm' is always +ve and |m| < 1.

■ Some important points related to lenses

● Convex lens :

- ▶ u = always negative.
- ▶ v = -ve, when object is placed between optical centre(O) and focus(F_1).
- ▶ v = +ve, for all other possible cases (real and inverted).
- ▶ f = +ve, power (P) = +ve.

● Concave lens :

- ▶ u = always negative.
- ▶ v = always -ve (virtual and erect).
- ▶ f = -ve, power (P) = -ve.
- ▶ Image always diminished.

■ Power of Lens : A lens of short focal length bends the light rays more, through large angles.

- The power of a lens is a measure of the degree of convergence or divergence of light rays falling on it
- The power of a lens is defined as the '**reciprocal of its focal length**'.

$$P = \frac{1}{f}$$

- **Unit of power of lens : Diopter (D)** 1 diopter = 1 m⁻¹

► The power of convex lens (converging lens) is '**positive**' and power of concave lens (diverging lens) is '**negative**'.

- **Two lenses in contact (lens combination)** : (a) $P = P_1 + P_2$ (b) $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$ (c) $m = m_1 \times m_2$

- **Example** : A concave lens has focal length of 15 cm. At what distance should the object from the lens be placed so that it forms an image at 10 cm from the lens ? Also, find the magnification produced by the lens.

Solution : A concave lens always forms a virtual, erect image on the same side of the object.

Image distance $v = -10$ cm ; focal length $f = -15$ cm ; object distance $u = ?$

Lens equation,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad \frac{1}{(-10)} - \frac{1}{u} = \frac{1}{(-15)} \quad \text{or} \quad \frac{1}{u} = \frac{1}{15} - \frac{1}{10} = \frac{2-3}{30} = \frac{-1}{30} \quad \text{or} \quad u = -30 \text{ cm}$$

$$\text{Magnification, } m = \frac{v}{u} = \frac{(-10)}{(-30)} = \frac{1}{3} = +0.33$$

The positive sign shows that the image is erect and virtual. The image is one-third of the size of the object.

- **Example** : A 2.0 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 10 cm. The distance of the object from the lens is 15 cm. Find the nature, position and size of the image. Also find its magnification.

Solution : Given, height of the object $h_1 = +2.0$ cm ; focal length $f = +10$ cm ; object distance $u = -15$ cm ; image-distance $v = ?$; height of the image $h_2 = ?$

Lens equation,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad \frac{1}{v} - \frac{1}{(-15)} = \frac{1}{(+10)} \quad \text{or} \quad \frac{1}{v} = \frac{1}{10} - \frac{1}{15} = \frac{3-2}{30} = \frac{+1}{30} \quad \text{or} \quad v = +30 \text{ cm}$$

$$\text{Magnification, } m = \frac{h_2}{h_1} = \frac{v}{u} \quad \text{or} \quad h_2 = \frac{v}{u} \times h_1 = \frac{+30}{-15} \times (+2) = -4 \text{ cm}$$

The negative sign of h_2 shows that the image is inverted and real. A real, inverted image, 4 cm tall, is formed at a distance of 30 cm on the other side of the lens.

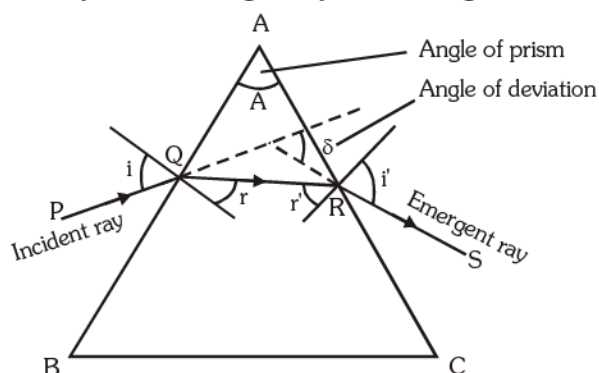
■ Refraction of light through a prism

Prism : It is a block of glass or transparent material, usually bounded by two triangular plane surface (base) and three lateral rectangular surfaces.

- For a parallel refracting surfaces, like in a glass slab, the emergent ray is parallel to the incident ray. In a triangular prism, there are two triangular base and three rectangular lateral surfaces. These surfaces are inclined to each other at some angle. The angle between its two lateral surfaces is known as the **angle of the prism** or **refracting angle**.

Let the incident ray is PQ (see fig.), QR is the refracted ray and RS is the emergent ray. The ray PQ enters from air to glass at the first refracting surface AB. The light ray after refraction bends towards the normal (refracted ray QR). At the second surface AC, the ray QR enters from glass to air and bends away from the normal after refraction. The emergent ray RS is not parallel to the incident ray PQ due to the peculiar shape of prism.

- The angle between the incident ray and the emergent ray is called 'angle of deviation (δ)'.



Formulae related to refraction of light through prism :

$$(1) \angle A = r + r'$$

$$(2) \delta = (i + i') - (r + r')$$

$$(3) \delta = i + i' - A$$

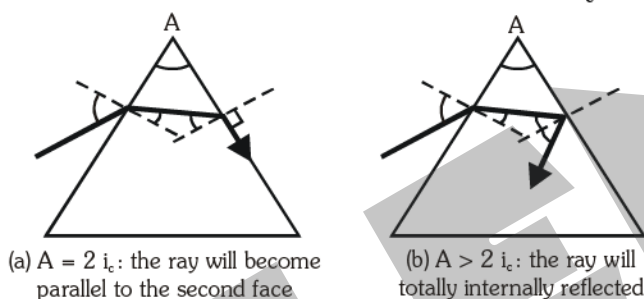
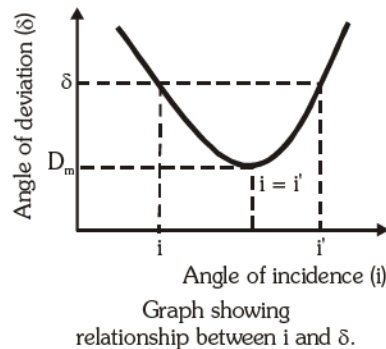
- The angle of deviation is minimum when $\angle i = \angle i'$ and $\angle r = \angle r'$.

$$n_{21} = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

- If A and D_m are quite small (i.e., thin prism), then above equation becomes,

$$D_m = (n_{21} - 1) A$$

- **Condition for no emergence :** If angle of the prism A is greater than or equal to twice the critical angle, the ray will not emerge from the second face of the prism (see fig.) i.e., $A \geq 2 i_c$.



Example : A ray fall on a prism ABC ($AB = BC$) and travel as shown in fig.52. What should be the least value of refractive index of material of the prism ?

Solution :

Since $AB = BC$, therefore, $\angle BAC = \angle CAB = x$ (let)

$$\angle BAC + \angle CAB + \angle ABC = 180^\circ \quad (\angle \text{sum property of a } \Delta)$$

$$\text{or } x + x + 90^\circ = 180^\circ$$

$$\text{or } 2x = 90^\circ \quad \text{or} \quad x = 45^\circ$$

From the fig. 52, it is clear that internal reflection is taking place at both faces AB and BC . Angle of incidences at both faces are 45° each.

Thus, critical angle, $i_c \leq 45^\circ$

$$\text{or } \sin i_c \leq \sin 45^\circ$$

$$\text{or } \sin i_c \leq \frac{1}{\sqrt{2}} \quad \left(\because \sin 45^\circ = \frac{1}{\sqrt{2}} \right) \text{ ---- (1)}$$

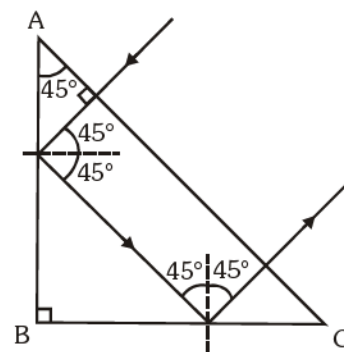
$$\text{Now, } \sin i_c = \frac{1}{n} \text{ ---- (2)}$$

Using (1) & (2), we get,

$$\frac{1}{n} \leq \frac{1}{\sqrt{2}}$$

$$\text{or } n \geq \sqrt{2}$$

Thus, the least possible value of refractive index of material of prism should be $\sqrt{2}$.



Example : The angles of a prism are 30° , 60° and 90° . What should be the refractive index of a liquid to be kept in contact with the longest face so that a ray normally incident on the smallest face gets just internally reflected at the longest ? ($n_{\text{prism}} = 1.5$)

Solution :

Let ΔABC is right angled triangle, with $\angle A = 30^\circ$, $\angle B = 90^\circ$, $\angle C = 60^\circ$
We know that shortest side in a triangle is opposite to the shortest angle, thus shortest side in the ΔABC is BC . Similarly, longest side is opposite to the largest angle, thus, longest side is AC .

Let refractive index of liquid in contact with AC be n_2 and refractive index of material of glass prism be n_1 .

Given, $n_1 = 1.5$

Now, an incident ray falls normally on the face BC and it gets refracted without any deviation, that is, it goes undeviated.

When it strikes the longest side, it just gets internally reflected. This means the angle of incidence is equal to the critical angle. That is,

$$i = i_c \text{ ---- (1)}$$

In ΔDEC , $\angle DCE + \angle CDE + \angle DEC = 180^\circ$ ($\angle$ sum property of a Δ)
or $60^\circ + \angle CDE + 90^\circ = 180^\circ$ or $\angle CDE = 30^\circ$ ---- (2)

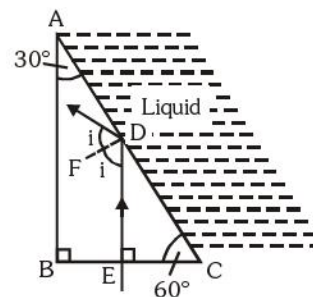
Now, $\angle EDF + \angle CDE = 90^\circ$ ($\because DF$ is $\perp$ to AC)

or $\angle EDF + 30^\circ = 90^\circ$ or $\angle EDF = 60^\circ$

or angle of incidence on face AC , $i = \angle EDF = 60^\circ$ ---- (3)

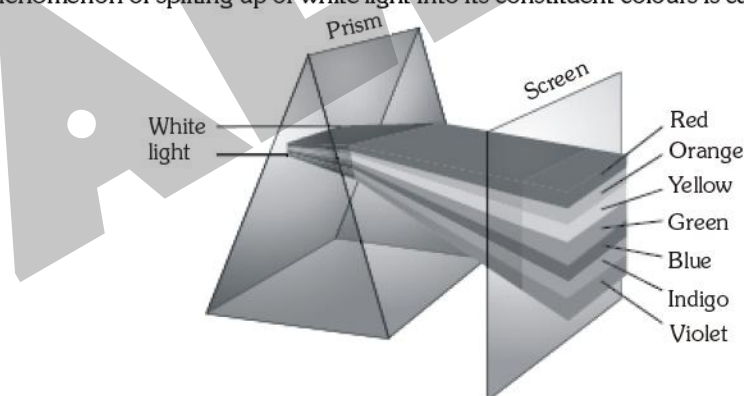
Using (1) and (3), we get, $i_c = 60^\circ$

$$\text{Now, } \sin i_c = \frac{1}{n_2} = \frac{n_2}{n_1} \quad \text{or} \quad \sin 60^\circ = \frac{n_2}{(1.5)} \quad \text{or} \quad n_2 = 1.5 \times \sin 60^\circ = \frac{3}{2} \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}$$

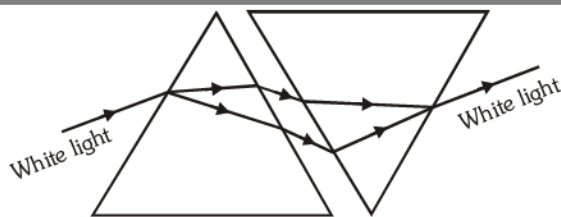


- **Dispersion of white light by a Prism :** If a beam of white light is made to fall on one face of a prism, the light emerging out from the other face of prism consists of seven colours namely violet (V), Indigo (I), Blue (B), Green (G), Yellow (Y), Orange (O) and Red (R).

- The deviation suffered by the red light is minimum and for the violet light it is maximum.
- The phenomenon of splitting up of white light into its constituent colours is called '**dispersion of light**'.



- **Reason :** The wavelengths of different colours of light are different and the refractive index of glass is different for different wavelengths. Higher the wavelength, lower will be the refractive index and thus, lower will be the deviation and vice-versa. Thus, deviation of red light is low as its wavelength is large. Similarly, deviation of violet light is high as its wavelength is small.
- Initially, it was a matter of debate, whether the prism itself create colour in some way or it only separate the colours already present in white light. Isaac Newton settled the issue by performing a simple experiment. He put another similar experiment. He put another similar prism, but in an inverted position, and let the emergent separated beam fall on the second prism. The resulting emergent beam was found to be white light. The first prism separated the white light into its component colours, which were recombined by the inverted prism to give white light. Thus, white light itself consists of colours which can be separated by the prism.



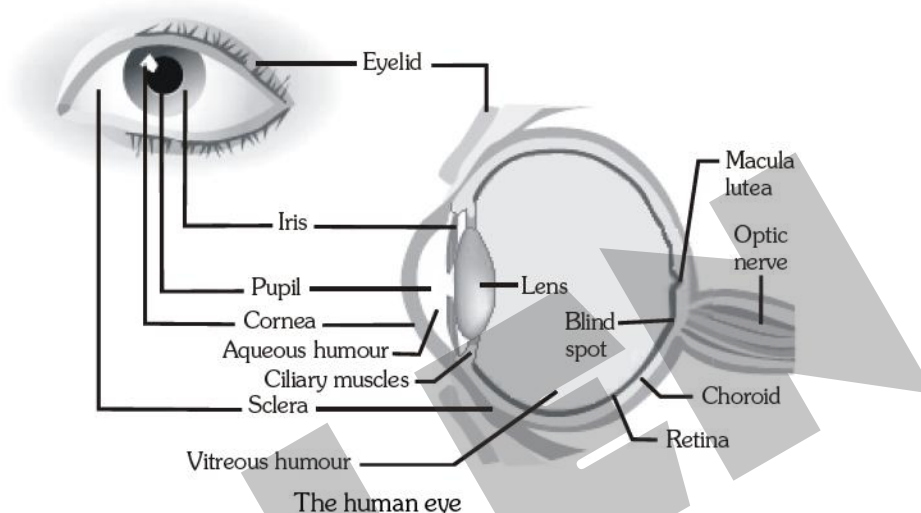
Newton's classic experiment on dispersion of white light

- ▶ Formation of **rainbow** involves the phenomenon of dispersion of white light.
- **Scattering of light** : The process in which the light rays are deflected by the particles of the medium through which they pass is called **scattering of light**.
 - **Tyndall effect** : The scattering of light as it passes through a medium containing small particles is called is called **Tyndall effect**.
 - ▶ When rays of light falls on the fine particles of a colloidal solution, the path of the beam is visible due to the scattering of light by the colloidal particles. This phenomenon is called 'Tyndall effect'. This phenomenon is observed when a fine beam of sunlight enters a room through a small hole and the light scatters from the smoke or dust particles present in the room atmosphere. Tyndall effect can also be seen when sunlight passes through a canopy of a dense forest. Here, tiny water droplets present in the misty air scatter the light.
 - ▶ The colour of the scattered light depends on the size of the scattering particles. Very fine particles scatter shorter wavelengths mainly blue light while particles of bigger size scatter light having longer wavelengths. If the size of particles is quite large, then, the scatter light is almost white.
 - **Why sky appears blue ?**

The molecules of air and other fine particles in the atmosphere have size quite smaller as compared to the wavelength of the visible light. These particles scatter the light having shorter wavelengths at the violet or blue end more than the light having longer wavelengths at the red end. When the sun light passes through the earth's atmosphere, the fine particles present in air scatter the blue colour more strongly in comparison to the red colour. The scatter blue light falls on the our eyes. Thus, the sky appears blue.

 - ▶ If the earth had no atmosphere, the would not have any scattering of light. Then, the sky would have appeared dark or black. The sky appears dark to the persons flying in the aeroplanes at very high altitudes, as scattering is not prominent at these heights.
 - ▶ The 'danger' signals are made of red colour because red colour is scattered least by the air particles, smoke or fog due to its longer wavelength. Hence, red colour can be seen in the same shade even at the large distance.
 - ▶ **Colour of the sun at sunrise and sunset** : At the sunrise or sunset, the sun is near the horizon. Light reaching the eye of an observer travels a larger distance. Thus, most of the blue light and shorter wavelength rays are scattered away by the air particles. Hence, the light that reaches the eye of an observer is of longer wavelengths (of red end). Thus, the sun appears reddish at sunrise or sunset.
 - ▶ At the noon the light has to travel a comparatively shorter distance before reaching the eye of an observer. Thus, only a very little amount of blue or violet light is scattered away. Hence, the sun at noon looks almost white.
- **The human eye**
 - **Components and structure** :
 - (a) The eye has a nearly spherical shape, the eye ball has a diameter of about one inch (nearly 2.3 cm). The front portion is more sharply curved and it is covered by a thin transparent protective membrane called 'cornea'. It is this portion which is visible from the outside.
 - (b) Behind cornea, there is a liquid called 'aqueous humour' and behind that, there is a crystalline lens. Between the aqueous humour and the lens, there is a muscular diaphragm called 'iris', which has a small hole in it called 'pupil'.
 - (c) The eye lens is composed of a fibrous, jelly like material which is hard in the middle and gradually becomes soft towards the edges. The curvature of the lens is altered by the ciliary muscles to which it is attached.
 - (d) The space between lens and the retina is filled with another liquid called 'vitreous humour'.

- (e) The light entering the eye forms an image on the retina which is a delicate membrane having enormous no of light sensitive cells. It contain about 125 million receptors called 'rod' and 'cones' which receive the light rays and about one million optic nerve fibre which transmit the information to the brain.
- (f) The region on the retina where the optic nerve enters the eye ball is called the 'blind spot'. It is insensitive to the light, that is, any image formed at this spot is not visible.
- (g) The macula lutea also called 'yellow spot' is a central part of the retina responsible for sensing fine detail and for looking straight ahead. It has high concentration of nerve endings and it is slightly raised. Its function is to form a very clear image, by sensing a large number of electrical signals to the brain
- (h) Sclera or sclerotic is the outermost covering of eye and it is made of white fibrous tissue. 'Choroid' is a grey membrane attached to sclera.



● Role of iris

Iris controls the size of the pupil and therefore, helps in regulating the amount of light entering the eye through a variable aperture (the pupil). In low intensity of light, iris expands the pupil to allow more light to enter into the eye. When the light is very bright, iris contracts the pupil and the pupil becomes very small, thus, only a small amount of light enter into the eye.

● Working of eye

- (a) When the light enters the eye from air, most of the bending of light occurs at cornea. Some additional bending is done by the lens so as to form an inverted, real image of the object on retina.
- (b) When the eye is focused on a distant object, the ciliary muscle relaxes allowing ligaments to increase tension on the lens and cause it to flatten i.e., the lens becomes thin or less curved. Finally, the image is formed on the retina and we see the object clearly.
- (c) When the eye is focused on a closer object, the ciliary muscle contracts, allowing the lens, by virtue of its elasticity, to become more curved i.e., the lens becomes thick. Again, the image is formed on the retina and we see the object clearly.
- (d) The light-sensitive cells get activated upon illumination and generate electrical signals. These signals are sent to the brain via the optic nerves. The brain interprets these signals, and finally, processes the information so that we perceive objects as they are i.e., the brain makes the inverted image formed on the retina again erect (or upright).
- The change in eye lens curvature changes its focal length. When the ciliary muscles are relaxed, the focal length of the eye lens is more (about 2.5 cm). The lens is then thin and we can see distant object clearly. When we look at nearby objects closer to eye, the ciliary muscles contract, so that the eye lens becomes more rounded in shape. The focal length of the eye lens decreases so as to bring light from the nearby objects into focus on the retina.
- **Accommodation :** By contracting or relaxing the ciliary muscles connected to the lens, its shape can be changed such that we can see the nearby as well as the distant objects clearly. This process is called accommodation.
 - ▶ The ability or property of the eye to change the shape of lens so as to see the object clearly is called 'accommodation'.

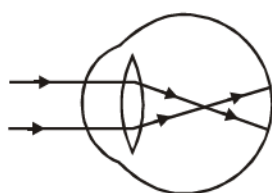
- ▶ The muscles cannot be strained beyond a limit and thus, an object placed too close to the eye cannot be seen clearly.
- **Near point** : The nearest point for which the image can be formed clearly on the retina, is called the 'near point of the eye'.
- ▶ **Least distance of distinct vision** : The minimum distance at which objects can be seen clearly without strain is called 'least distance of distinct vision (or clear vision)'. In other words, the distance of the near point from the eye is called the 'least distance of distinct vision'.
- ▶ This distance varied with age, increases with it because of decreasing effectiveness of the ciliary muscles and the loss of flexibility of the lens. The symbol used for least distance of distinct vision is D . Standard value of D for a young adult with normal vision is 25 cm. For a child of 10 years D is nearly 7 to 8 cm, for a old man of 60 years D is nearly 200 cm.
- **Far Point** : The farthest point up to which the eye can see objects clearly is called the 'far point' of the eye. For normal eye, far point is at infinity.
- **Persistence of Vision** : The continuance of the sensation of the eye for a fraction of second even after the removal of the object is called '**persistence of vision**'.
- ▶ **Principle of cinematography or motion-picture projection** : If still pictures taken by a movie camera are projected on a screen at a rate of about 24 images per second, then the image of one picture persists on the retina till the next picture falls on the screen. Due to this, the slightly different images of the successive pictures present on the film merge smoothly into one another and gives us the feeling of moving images (or movie). This is the principle used in cinematography.
- **How do we see colours ?**
Our retina has a large no. of light sensitive cells having shapes of rods and cones. The rod shaped cells respond to the intensity of light with different degree of brightness and darkness. In dim light, the rods are sensitive but the cones are not. But, the rods cannot distinguish between various colours. It is the cone-shaped cells which respond to colours. They are active only in bright light and make colour perception possible. The cones are sensitive to red, green and blue light to different extents. When a particular colour like red falls on the retina, it mainly stimulates the red colour sensitive cones than the other kinds of cones.
- **Colour blindness**
Some people do not possess some cone cells that respond to certain specific colours only. This is a genetic disorder called '**colour blindness**'. The persons who cannot distinguish between colours but can see well are called '**colour blind**'.
- ▶ The defect of eye due to which a person cannot distinguish between colours is called '**colour blindness**'. Driving license for vehicles is not issued to colour blind persons. Usually, colour blind persons cannot distinguish between red and green colours.
- Colour perception of different animals is different due to different structure of rods and cones:
 - ▶ Bees has some retina cones which are sensitive to ultra-violet rays, which have wavelengths shorter than that of the violet light. Thus, Bees can see objects in ultra-violet rays and can recognise colours.
 - ▶ Human beings cannot see objects in ultra-violet rays as their retina cones are not sensitive to such short wavelengths that are possessed by ultra-violet rays.
 - ▶ The retina of a chick has mostly cones and only a few rods. As the cones are sensitive to bright light only, the chicks wakes up with sunrise (dawn) and sleep by the sunset (dusk).
- **Cataract** : The crystalline lens of some people in old age becomes hazy or even opaque due to development of membrane over it. This defect is called '**cataract**' which leads to decrease or loss of vision of the eye.
 - ▶ Cataract can be removed by performing surgery to restore clear vision.

■ Defects of Visions

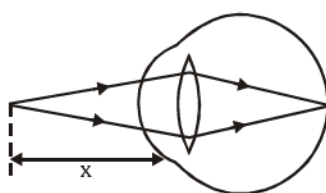
- **Myopia (Near sightedness)** : A person suffering from this defect cannot see distant objects clearly. This is because the maximum focal length is less than distance between the lens and the retina. The parallel rays coming from the distant object focus short of the retina. The ciliary muscles are fully relaxed in this case and any strain in it can only further decrease the focal length which is of no help to see distant objects.
 - ▶ **Reason** : This defect arises because the power of eye lens is too great due to the decrease in focal length of the eye lens. This may arise due to either (1) excessive curvature of the cornea. (2) elongation of the eye ball.
 - ▶ **Correction** : Myopia is corrected by using an appropriate concave lens (diverging lens) which increases the focal length in order to bring the image of the object back on the retina itself.

- ▶ Let a person can see clearly to a distance 'x' only. For myopia, concave lens is used, whose focal length is given by,

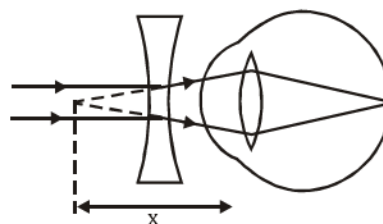
$$f = -x \quad \text{and} \quad \text{power, } P = \frac{1}{f} = \frac{-1}{x} \quad ('x' \text{ is always taken +ve, 'x' and 'f' are in meters})$$



A myopic eye



Far point of a myopic eye

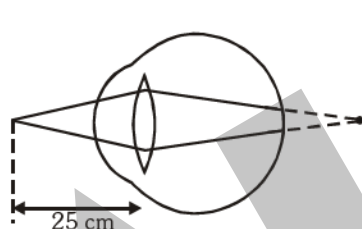


Correction for a myopic eye

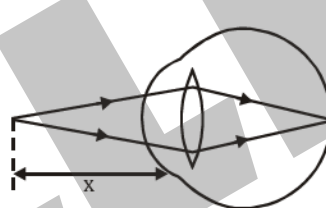
● Hypermetropia (far-sightedness)

A person suffering from Hypermetropia cannot see clearly the objects closer to the eye. The least distance of distinct vision is quite larger than 25 cm for that person and the person has to keep the object inconveniently away from the eye. Thus, the image is not formed at the retina if an object is kept at about 25 cm away from the eye. The rays are focused behind the retina.

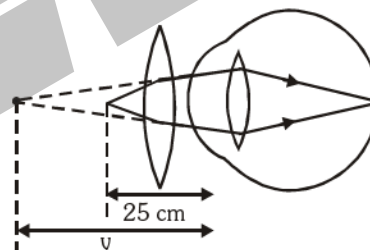
- ▶ **Reason :** This defect arises because either the focal length of the eye lens is too great or eye ball becomes too short. Due to this, light rays from nearby objects cannot be brought to focus on the retina to give a clear image.
- ▶ **Correction :** Hypermetropia can be corrected by using a convex lens (converging lens) which decreases the focal length in order to focus the image of a nearby object at the retina. Thus, a person can see the object clearly. Let 'y' be the minimum distance at which the person can see the object clearly.



A hypermetropic eye



Near point for a hypermetropic eye



Correction for a hypermetropic eye

- ▶ For hypermetropia, convex lens is used, whose focal length is calculated by following formula,

$$\frac{1}{f} = \frac{1}{0.25} - \frac{1}{y} \quad \text{and} \quad \text{power, } P = \frac{1}{f} = \frac{1}{0.25} - \frac{1}{y} \quad ('y' \text{ is always taken +ve, 'y' and 'f' are in meters})$$

● Presbyopia

The defect that arises due to ageing in which a person cannot read comfortably and distinctly without corrective eye glasses is called '**presbyopia**'.

- ▶ **Reason :** The power of accommodation of the eye decreases with ageing. For most of the people the near point recedes, this means, the least distance of distinct vision increases. This phenomenon arises due to the gradual weakening of ciliary muscles and decreasing flexibility of the crystalline eye lens.
- ▶ **Correction :** Use of convex lens.
- Some times, a person may suffer from both myopia and hypermetropia. Such people often use **bi-focal lens**. Commonly, the upper portion of bifocal lens is a concave lens (used for distant vision) and the lower portion is a convex lens (used for reading purpose).
- **Astigmatism :** In astigmatism, a person cannot see all the directions equally well. This means objects in one direction being well-focused while those in other direction are not well-focused on the retina.
- ▶ **Reason :** This defect usually arises due to cornea which is not perfect spherical. Due to this, it has different curvatures in different directions in vertical and horizontal planes. This defect also arises due to defective eye lens which has different curvatures along different directions.

- **Correction** : This defect can be corrected by using lenses with different curvatures in different planes which compensate for irregularities in the cornea (or eye lens) called '**cylindrical lenses**'.

- These days, it is possible to correct the defects of vision by using '**contact lenses**'.

■ Microscope

It is an optical instrument used for viewing tiny objects clearly, by magnifying the size of image formed at the retina.

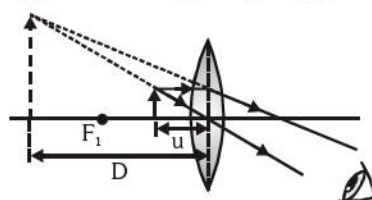
Two essential component in design of optical instruments for viewing an object distinctly are :

- (1) Greater (or larger) magnification
- (2) Adequate (or sufficient) resolution.

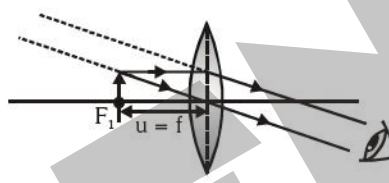
- Resolution of an optical instrument is its ability to resolve two closely spaced point objects, which can be distinguished as separate.

Simple microscope

A simple microscope is a converging lens (or convex lens) of short focal length. The lens is held near the object at a distance less than the focal length or exactly at focal length. The eye is positioned close to the lens on the other side. An erect, magnified and virtual image of the object is formed. The image can be formed at near point [see fig.54(a)] or at infinity [see fig.54(b)].



(a) Eye focused at near point



(b) Eye focused at infinity

Formulae for magnification in simple microscope

- (1) Eye focused at near point :

$$m = \left(1 + \frac{D}{f}\right) \quad (D \text{ and } f \text{ are positive})$$

- (2) Eye focused at infinity :

$$m = \frac{D}{f} \quad (D \text{ and } f \text{ are positive})$$

Compound microscope

In a compound microscope, there are two lenses, one compounding the effect of other.

- The lens near the object is called the '**objective**'. The objective (O) is a convex lens of short focal length and smaller aperture (size). It is mounted in a hollow, long, cylindrical metal tube.
- The lens near the eye is called '**eyepiece**' or '**ocular**'. It is a convex lens of larger focal length and larger aperture (size) than the objective. The eyepiece is also mounted in a metal tube, which can slide inside a bigger metal tube.

Formulae for magnification in compound microscope :

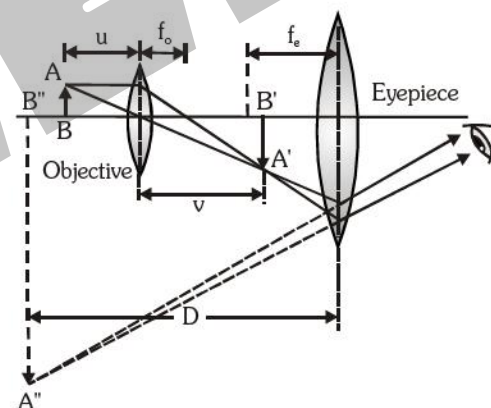
- (1) Eye focused at near point : $m = m_e \times m_o$

$$\text{or } m = \left(1 + \frac{D}{f_e}\right) \left(\frac{v}{f_o} - 1\right) \quad (f_e, f_o, v, D \text{ all are positive})$$

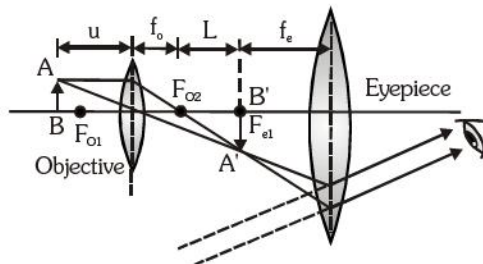
- (2) Eye focused at infinity : $m = m_e \times m_o$ or $m = \frac{L D}{f_o f_e}$

Where, L is the distance between second focal point (F_{o2}) of the objective and the first focal point (F_{e1}) of the eyepiece and it is called '**tube length**' of the compound microscope.

Distance between lenses = $L + f_{o2} + f_{e1}$



(a) Eye focused at near point



(b) Eye focused at infinity

Compound microscope

Example : A converging lens of focal length 6.25 cm is used as a magnifying glass. If near point of the observer is 25 cm from the eye and lens is held close to eye, calculate (i) the distance of the object from the lens (ii) the magnification (iii) Also, find the magnification when the final image is formed at infinity.

Solution :

(i) $f = + 6.25 \text{ cm}$, $v = - 25 \text{ cm}$, $u = ?$

Lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad \frac{1}{(-25)} - \frac{1}{u} = \frac{1}{(+6.25)}$$

$$\text{or} \quad \frac{1}{u} = -\frac{1}{25} - \frac{1}{6.25} = \frac{-1-4}{25} = \frac{-5}{25} = -\frac{1}{5}$$

$$\text{or} \quad u = - 5 \text{ cm}$$

(ii) $m = 1 + \frac{D}{f} = 1 + \frac{25}{6.25} = 5$

(iii) When the image is formed at the infinity,

$$m = \frac{D}{f} = \frac{25}{6.25} = 4$$

■ Telescopes

Telescopes are used to provide magnification of distant objects.

Refracting astronomical telescope

It is an optical instrument used to see the magnified image of celestial objects like stars, planets, etc. and it makes use of lenses only.

Construction : It consists of two lenses, an objective O and an eye piece E, mounted co-axially.

- The lens used to collect the light from the distant object is called '**objective**'. It is a convex lens of large focal length and a larger aperture than the eye piece.
 - ▶ The objective has a larger aperture in order to collect large amount of light of distant objects and thus, forms bright images.
- The lens held near to the eye is called '**eyepiece**'. It is a convex lens of small focal length and small aperture so that the whole light inside the telescope tube may enter into the eye for clear vision.

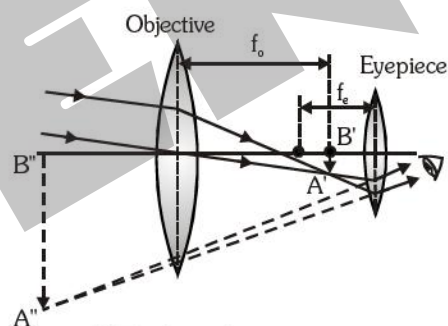
Formulae for magnification in telescope :

(1) Eye focused at near point : $m = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$

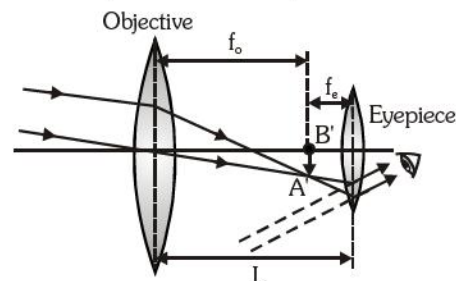
(2) Eye focused at infinity : $m = \frac{f_o}{f_e}$; $L = f_o + f_e$

In this case, distance L is the distance between the optical centres of the objective and eyepiece when the image is formed at infinity and it is called '**length of the telescope tube**'.

- The telescope is said to be in '**normal adjustment**' when the image is formed at infinity and the distance between the objective and the eye piece is equal to the sum of their focal lengths. Eye is relaxed in this case as it is focused at infinity.



(a) Eye focused at near point



(b) Eye focused at infinity

Refracting astronomical telescope